

Spinors, Dirac Equation, CPT

Source: *M. D. Schwartz, Quantum Field Theory and the Standard Model, Chapters 10–11.*

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Representations of the Lorentz Group

We have tried to find unitary irreducible representations of the Poincaré group by writing down a Lagrangian with positive definite energy. The next logical step would be to find a way to systematically embed particles into fields. By doing this, we will reveal the existence of spin- $\frac{1}{2}$ particles.

Group Theory

A group is a set of elements $\{g_i\}$ and a rule $g_i \times g_j = g_k$ which tells how each set of elements is multiplied to get a third. More precisely, the rule needs to be associative, and there needs to be an identity group element and inverses. A representation is a particular embedding of these g_i into operators that act on a vector space. Any group has the trivial representation $r : g_i \rightarrow \mathbb{1}$. A representation in which each group element gets into its own matrix is called a faithful representation. Recall Lorentz group preserves the Minkowski metric

$$\Lambda^T g \Lambda = g. \quad (1.1)$$

The matrices in this equation are in 4-vector representation. Examples of Lorentz transformations are rotations around the x , y or z axes:

$$\begin{pmatrix} 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & \cos \theta_x & \sin \theta_x \\ 0 & 0 & -\sin \theta_x & \cos \theta_x \end{pmatrix}, \quad \begin{pmatrix} 1 & 0 & 0 & 0 \\ 0 & \cos \theta_y & 0 & -\sin \theta_y \\ 0 & 0 & 1 & 0 \\ 0 & \sin \theta_y & 0 & \cos \theta_y \end{pmatrix}, \quad \begin{pmatrix} 1 & 0 & 0 & 0 \\ 0 & \cos \theta_z & \sin \theta_z & 0 \\ 0 & -\sin \theta_z & \cos \theta_z & 0 \\ 0 & 0 & 0 & 1 \end{pmatrix}$$

and boosts in the x , y or z directions:

$$\begin{pmatrix} \cosh \beta_x & \sinh \beta_x & 0 & 0 \\ \sinh \beta_x & \cosh \beta_x & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \end{pmatrix}, \quad \begin{pmatrix} \cosh \beta_y & 0 & \sinh \beta_y & 0 \\ 0 & 1 & 0 & 0 \\ \sinh \beta_y & 0 & \cosh \beta_y & 0 \\ 0 & 0 & 0 & 1 \end{pmatrix}, \quad \begin{pmatrix} \cosh \beta_z & 0 & 0 & \sinh \beta_z \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 1 & 0 \\ \sinh \beta_z & 0 & 0 & \cosh \beta_z \end{pmatrix}.$$

These matrices give an embedding of elements of the Lorentz group into a set of matrices. That is, they describe one particular representation of the Lorentz group (the 4-vector representation). We would now like to find all the representations.

To extract the group away from its representation, it is easiest to look at the infinitesimal transformations. In the 4-vector representation:

$$\delta X_0 = \beta_i X_i, \quad (1.2)$$

$$\delta X_i = \beta_i X_i - \varepsilon_{ijk} \theta_j X_k. \quad (1.3)$$

Generally, the infinitesimal transformations can be written as

$$\delta X_\mu = i \left[\sum_{i=1}^3 \theta_i (J_i)_{\mu\nu} + \beta_i (K_i)_{\mu\nu} \right] X_\nu, \quad (1.4)$$

where

$$J_1 = i \begin{pmatrix} 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & -1 \\ 0 & 0 & 1 & 0 \end{pmatrix}, \quad J_2 = i \begin{pmatrix} 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 1 \\ 0 & 0 & 0 & 0 \\ 0 & -1 & 0 & 0 \end{pmatrix}, \quad J_3 = i \begin{pmatrix} 0 & 0 & 0 & 0 \\ 0 & 0 & -1 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 0 & 0 \end{pmatrix},$$

$$K_1 = i \begin{pmatrix} 0 & -1 & 0 & 0 \\ -1 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 \end{pmatrix}, \quad K_2 = i \begin{pmatrix} 0 & 0 & -1 & 0 \\ 0 & 0 & 0 & 0 \\ -1 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 \end{pmatrix}, \quad K_3 = i \begin{pmatrix} 0 & 0 & 0 & -1 \\ 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 \\ -1 & 0 & 0 & 0 \end{pmatrix}.$$

These matrices are known as the generators of the Lorentz group in the 4-vector basis. They generate the group in the sense that any element of the group can be written uniquely as

$$\Lambda = \exp(i\theta_i J_i + i\beta_i K_i) \quad (1.5)$$

up to some discrete transformations. The advantage of writing the group elements this way is that it is completely general. In any finite-dimensional representation the group elements can be written as an exponential of matrices.

For any group G , some group elements $g \in G$ can be written as $g = \exp(ic_i^g \lambda_i)$, where c_i^g are numbers and λ_i are group generators. The generators are in an algebra, because you can add and multiply them, while the group elements are in a group, because you can only multiply them. For example, the real numbers form an algebra (there is a rule for addition and a rule for multiplication) but rotations are a group (there is only one rule, multiplication).

Lie groups are a class of groups, including the Lorentz group, with an infinite number of elements but a finite number of generators. The generators of the Lie group form an algebra called its Lie algebra. Lie groups are critical to understanding the Standard Model, since QED is described by the unitary group $U(1)$, the weak force by the special unitary group $SU(2)$ and the strong force by the group $SU(3)$. The Lorentz group is sometimes called $O(1, 3)$. This is an orthogonal (preserves a metric) group corresponding to a metric with $(1, 3)$ signature (i.e. $g_{\mu\nu} = \text{diag}(1, -1, -1, -1)$).

Lie groups also have the structure of a differentiable manifold. For most applications of quantum field theory, the manifold is totally irrelevant, but it is occasionally important. For example, topological properties of the 3D rotation group $SO(3)$ will help us understand the spin-statistics theorem. We sometimes distinguish the proper orthochronous Lorentz group, which is the elements of the Lorentz group continuously connected to the identity, from the full Lorentz group, which includes time reversal (T) and parity reversal (P).

In a Lie algebra, the multiplication rule is defined as the Lie bracket. With matrix representations, this Lie bracket is just an ordinary commutator. Since any element of the Lie algebra can be represented as a linear combination of its generators. For the Lorentz group, these commutation relations can be calculated with any representation, including the 4-vector representation. Thus,

$$[J_i, J_j] = i\epsilon_{ijk}J_k, \quad (1.6)$$

$$[J_i, K_j] = i\epsilon_{ijk}K_k, \quad (1.7)$$

$$[K_i, K_j] = -i\epsilon_{ijk}J_k. \quad (1.8)$$

These commutation relations define the Lorentz algebra, $so(1,3)$. Notice that $[J_i, J_j] = i\epsilon_{ijk}J_k$ is the algebra for rotations, $SO(3)$, and in fact the J_i generate the 3D rotation subgroup of the Lorentz group.

These commutation relations define the Lie algebra of the Lorentz group. Although these commutation relations were derived using Eq. (10.14) they must hold for any representation; for example in the rank-2 tensor representation J_i and K_i can be written as 16-dimensional matrices. It is sometimes useful to use a different form for these commutation relations. We can index the generators by $V^{\mu\nu}$ instead of J_i and K_i :

$$V^{\mu\nu} = \begin{pmatrix} 0 & K_1 & K_2 & K_3 \\ -K_1 & 0 & J_3 & -J_2 \\ -K_2 & -J_3 & 0 & J_1 \\ -K_3 & J_2 & -J_1 & 0 \end{pmatrix}. \quad (1.9)$$

Here each $V^{\mu\nu}$ is itself a 4×4 matrix, for example, $V^{23} = J_1$. A Lorentz transformation can be written in terms of $V^{\mu\nu}$ as $\Lambda_V = \exp(i\theta_{\mu\nu}V^{\mu\nu})$ for six numbers $\theta_{\mu\nu}$. These $V^{\mu\nu}$ satisfy

$$[V^{\mu\nu}, V^{\rho\sigma}] = i(g^{\nu\rho}V^{\mu\sigma} - g^{\mu\rho}V^{\nu\sigma} - g^{\nu\sigma}V^{\mu\rho} + g^{\mu\sigma}V^{\nu\rho}). \quad (1.10)$$

By definition, the generators in any other representation must satisfy these same relations. For example, another representation of the Lorentz group is given by

$$L^{\mu\nu} = i(x^\mu\partial^\nu - x^\nu\partial^\mu). \quad (1.11)$$

This is an infinite-dimensional representation which acts on functions rather than a finite-dimensional vector space. These are the classical generators of angular momentum generalised to include time. You can check that $L^{\mu\nu}$ satisfies the commutation relations of the Lorentz algebra.

By the way, not all the elements of the Lorentz group can be written as $\exp(ic_i^g \lambda_i)$ for some c_i^g . The generators of the Lorentz algebra $\mathfrak{so}(1, 3)$ only generate the part of the Lorentz group connected to the identity, known as the proper orthochronous Lorentz group $SO^+(1, 3)$. It is possible for two different groups to have the same algebra. For example, the proper orthochronous Lorentz group and the full Lorentz group have the same algebra, but the full Lorentz group has in addition time reversal and parity. The group generated by $\mathfrak{so}(1, 3)$ and T is called the orthochronous Lorentz group, denoted $O^+(1, 3)$. The proper Lorentz group is the special orthogonal group $SO(1, 3)$, which contains only the elements with determinant 1, so it excludes T and P . Sometimes $SO(1, 3)$ is taken to include only P with $SO^+(1, 3)$ excluding also T . These notations are more general than we need: in odd space-time dimensions, parity has determinant -1 and is therefore a special orthogonal transformation. Rather than worry about group naming conventions, we will simply talk about the Lorentz group with or without T and P .

General Representations of the Lorentz Group

Remember, the algebraic properties of the Lorentz group are independent of representations. Now, we can take linear combinations of J_i and K_i to construct

$$J_i^+ \equiv \frac{1}{2}(J_i + iK_i), \quad J_i^- \equiv \frac{1}{2}(J_i - iK_i). \quad (1.12)$$

These operators satisfy

$$[J_i^+, J_j^+] = i\epsilon_{ijk} J_k^+, \quad (1.13)$$

$$[J_i^-, J_j^-] = i\epsilon_{ijk} J_k^-, \quad (1.14)$$

$$[J_i^+, J_j^-] = 0. \quad (1.15)$$

These algebras are the 3D rotation algebra, $\mathfrak{so}(3) = \mathfrak{sl}(2, \mathbb{R}) = \mathfrak{so}(1, 1) = \mathfrak{su}(2)$ (it has multiple names because these Lie groups share the same algebra). Also, these two subalgebras commute. Therefore, we have shown that

$$\mathfrak{so}(1, 3) = \mathfrak{su}(2) \oplus \mathfrak{su}(2). \quad (1.16)$$

Thus, representations of $\mathfrak{su}(2) \oplus \mathfrak{su}(2)$ will determine representations of the Lorentz group.

We should be very familiar with $\mathfrak{su}(2)$. In quantum mechanics, the Pauli matrices satisfy exactly $\mathfrak{su}(2)$, which generate the 3D rotation group $SO(3)$. Each irreducible representation of $\mathfrak{su}(2)$ is characterised by a half-integer j . The representation acts on a vector space with $2j + 1$ basis elements.

Now, let us try to construct the finite-dimensional irreducible representations of $SU(2)$. By definition, such a representation is a set of three $n \times n$ matrices τ_1, τ_2 and τ_3 satisfying the algebra of the Pauli matrices $[\tau_i, \tau_j] = i\epsilon_{ijk} \tau_k$. Similarly, we can define the linear combination

$$\tau^\pm \equiv \tau^1 \pm i\tau^2. \quad (1.17)$$

In any representation we can diagonalize τ_3 . Its eigenvectors are n complex vectors V_j with $\tau_3 V_j = \lambda_j V_j$. We can write

$$\begin{aligned} \tau^3 \tau^\pm V_j &= ([\tau^3, \tau^\pm] + \tau^\pm \tau^3) V_j \\ &= (\pm \tau^\pm + \tau^\pm \tau^3) V_j \\ &= \tau^\pm (\tau^3 \pm 1) V_j \\ &= (\lambda_j \pm 1) (\tau^\pm V_j), \end{aligned} \quad (1.18)$$

which says that $\tau^\pm V_j$ are eigenvectors of τ^3 with eigenvalue $\lambda_j \pm 1$, except when $\tau^\pm V_j = 0$.

We can also prove that there must be one eigenstate $V_{\max}$ of τ^3 satisfying $\tau^+ V_{\max} = 0$. Similarly, there will be one eigenstate $V_{\min}$ of τ^3 satisfying $\tau^- V_{\min} = 0$. The proof goes as following. We can rank V_j from the one with the smallest eigenvalue to the one with the biggest, and thus we have a sequence of eigenvectors:

$$V_{\min}, V_2, V_3, \dots, V_{\max}. \quad (1.19)$$

From (1.18), we know that we can act τ^+ on V_j to take it to V_{j+1} . However, this cannot go forever, since there are only a finite number of V_j . Therefore, there must be a $V_{\max}$ such that $\tau^+ V_{\max} = 0$. Similarly, there must exist a $V_{\min}$ such that $\tau^- V_{\min} = 0$.

The eigenvalue $\lambda_{\max} = j$ is known as the spin. There exist an N such that $(\tau^-)^N V_{\max} = V_{\min}$, and thus $(\tau^-)^{N+1} V_{\max} = 0$. Hence, $\tau^+ (\tau^-)^{N+1} V_{\max} = 0$. Now, let us consider the general form

$$\tau^+ (\tau^-)^m V_{\max}, \quad (1.20)$$

where m is an integer. Using the commutation relations, we have

$$\begin{aligned} \tau^+ (\tau^-)^m V_{\max} &= \tau^- \tau^+ (\tau^-)^{m-1} V_{\max} + [\tau^+, \tau^-] (\tau^-)^{m-1} V_{\max} \\ &= [\tau^- \tau^+ (\tau^-)^{m-1} + 2\tau^3 (\tau^-)^{m-1}] V_{\max} \\ &= [\tau^- \tau^+ + 2(j - m + 1)] (\tau^-)^{m-1} V_{\max} \\ &= \tau^- [\tau^- \tau^+ + 2(j - m + 2)] (\tau^-)^{m-2} V_{\max} + 2(j - m + 1) (\tau^-)^{m-1} V_{\max} \\ &= (\tau^-)^2 \tau^+ (\tau^-)^{m-2} V_{\max} + 2[(j - m + 1) + (j - m + 2)] (\tau^-)^{m-1} V_{\max} \\ &\dots \\ &= (\tau^-)^m \tau^+ V_{\max} + 2[(j - (m - 1)) + (j - (m - 2)) + \dots \\ &\quad + (j - 1) + j] (\tau^-)^{m-1} V_{\max} \\ &= 2 \left[mj - \frac{m(m-1)}{2} \right] V_{\max} \\ &= m(2j - m + 1) V_{\max}. \end{aligned} \quad (1.21)$$

Therefore, we have

$$(\tau^-)^{N+1} V_{\max} = (N + 1)(2j - N) V_{\max} = 0, \quad (1.22)$$

which implies that

$$N = 2j. \quad (1.23)$$

Thus, we have $n = N + 1 = 2j + 1$. This says that the representation lives in a vector space with $2j + 1$ basis elements, which is exactly what we set out to prove. Since n is an integer, j must be a half-integer.

As the algebra of the Lorentz group is given by $\mathfrak{su}(2) \oplus \mathfrak{su}(2)$. Then, the irreducible representations of the Lorentz group should be characterised by two half-integers, A and B . The (A, B) representation has $(2A + 1)(2B + 1)$ degrees of freedom.

Since the 3D rotation group $\text{SO}(3)$ is a subgroup of the Lorentz group, every representation of the Lorentz group will also be a representation of $\text{SO}(3)$. In fact, finite-dimensional irreducible representations of the Lorentz algebra, generate many representations of $\text{SO}(3)$: with spins $j = A + B, A + B - 1, \dots, |A - B|$, as shown in Table Tab Group.

The relevance of the decomposition for particle physics is that Lagrangians are constructed out of fields, which transform under the Lorentz group. However, particles transform under irreducible unitary representations of the Poincaré group, which have spins associated with the little group. So, the decomposition of Lorentz representations determines the spins of particles that might be described by given fields.

By the way, the group generated by exponentiating the Lie algebra of a given group is known as the universal cover of the given group. For example, exponentiating $\mathfrak{su}(2)$ gives $\text{SU}(2)$. Since $\text{SU}(2)$ and $\text{SO}(3)$ have the same Lie algebra, $\text{SU}(2)$ is the universal cover of $\text{SO}(3)$. The Lie algebra $\mathfrak{su}(2) \oplus \mathfrak{su}(2)$ generates $\text{SL}(2, \mathbb{C})$, which is therefore the universal cover of the Lorentz group.

Spinor Representations

There are two complex $J = \frac{1}{2}$ representations: $(\frac{1}{2}, 0)$ and $(0, \frac{1}{2})$. The vector space they act on has $2J + 1 = 2$ degrees of freedom. Thus, we need to find 2×2 matrices that satisfy

$$[J_i^+, J_j^+] = i\epsilon_{ijk}J_k^+, \quad (2.1)$$

$$[J_i^-, J_j^-] = i\epsilon_{ijk}J_k^-, \quad (2.2)$$

$$[J_i^+, J_j^-] = 0. \quad (2.3)$$

Fortunately, we already knew such matrices, the Pauli matrices σ_i . The Pauli matrices have the commutation relation $[\sigma_i, \sigma_j] = 2i\epsilon_{ijk}\sigma_k$. Hence, we have

$$\left[\frac{\sigma_i}{2}, \frac{\sigma_j}{2} \right] = i\epsilon_{ijk} \frac{\sigma_k}{2}, \quad (2.4)$$

which is the $\text{SO}(3)$ algebra. Another useful identity is

$$\{\sigma_i, \sigma_j\} = 2\delta_{ij}. \quad (2.5)$$

To construct the $(\frac{1}{2}, 0)$ representation, we can set $J_i^- = \frac{\sigma_i}{2}$, which generates the $\frac{1}{2}$. For the $\mathbf{0}$, we can trivially set $J_i^+ = 0$. So the $(\frac{1}{2}, 0)$ representation is

$$\vec{J}^- = \frac{\vec{\sigma}}{2}, \quad \vec{J}^+ = 0. \quad (2.6)$$

Similarly, the $(0, \frac{1}{2})$ representation is given by

$$\vec{J}^- = 0, \quad \vec{J}^+ = \frac{\vec{\sigma}}{2}. \quad (2.7)$$

So what would the Lorentz transformation look like? Recall, $\vec{J} = \vec{J}^- + \vec{J}^+$ and $\vec{K} = i(\vec{K}^- - \vec{K}^+)$. Thus, we find

$$\left(\frac{1}{2}, 0\right) : \quad \vec{J} = \frac{\vec{\sigma}}{2}, \quad \vec{K} = i\frac{\vec{\sigma}}{2}, \quad (2.8)$$

$$\left(0, \frac{1}{2}\right) : \quad \vec{J} = \frac{\vec{\sigma}}{2}, \quad \vec{K} = -i\frac{\vec{\sigma}}{2}. \quad (2.9)$$

Since the Pauli matrices are Hermitian, $\vec{\sigma} = \vec{\sigma}^\dagger$, the rotations must also be Hermitian, and the boosts are anti-Hermitian, $\vec{K} = -\vec{K}^\dagger$.

The elements of vector space on which the spin- $\frac{1}{2}$ representation acts on is called the spinors. The $(0, \frac{1}{2})$ is known as the right-handed Weyl spinors and often denoted by ψ_R , while the $(\frac{1}{2}, 0)$ is known as the left-handed Weyl spinors and often denoted by ψ_L . Under rotation angles θ_j and boost angles β_j , these spinors transform as

$$\psi_R \rightarrow e^{\frac{1}{2}(i\theta_j\sigma_j + \beta_j\sigma_j)}\psi_R = \left(1 + \frac{i}{2}\theta_j\sigma_j + \frac{1}{2}\beta_j\sigma_j + \cdots\right)\psi_R, \quad (2.10)$$

$$\psi_L \rightarrow e^{\frac{1}{2}(i\theta_j\sigma_j - \beta_j\sigma_j)}\psi_L = \left(1 + \frac{i}{2}\theta_j\sigma_j - \frac{1}{2}\beta_j\sigma_j + \cdots\right)\psi_L. \quad (2.11)$$

For an infinitesimal transformation, we find

$$\delta\psi_R = \frac{1}{2}(i\theta_j + \beta_j)\sigma_j\psi_R, \quad (2.12)$$

$$\delta\psi_L = \frac{1}{2}(i\theta_j - \beta_j)\sigma_j\psi_L. \quad (2.13)$$

Similarly,

$$\delta\psi_R^\dagger = \frac{1}{2}(-i\theta_j + \beta_j)\psi_R^\dagger\sigma_j, \quad (2.14)$$

$$\delta\psi_L^\dagger = \frac{1}{2}(-i\theta_j - \beta_j)\psi_L^\dagger\sigma_j. \quad (2.15)$$

Note that we have non-trivial action for all the Lorentz generators, so these are the faithful irreducible representations of the Lorentz group.

Unitary Representations

Now, let us check if the 2D representations are unitary or not? Unitary means $\Lambda^\dagger \Lambda = 1$, which is the necessary condition for the matrix elements to be Lorentz-invariant:

$$\langle \psi | \psi \rangle \rightarrow \langle \psi | \Lambda^\dagger \Lambda | \psi \rangle. \quad (2.16)$$

SU(2) is the special unitary group, all of its representations are unitary. Since the group elements are given by $e^{i\Lambda}$, unitary requires Λ to be Hermitian. Thus, the generators for the SU(2) $\times$ SU(2) decomposition, $\vec{J}_\pm$, are Hermitian, which means $\exp(i\theta_j^+ J_j^+ + i\theta_j^- J_j^-)$ is unitary for $\theta_i^\pm \in \mathbb{R}$. However, this does not imply that the corresponding Lorentz group are unitary. The rotation angles θ_i and boost angles β_i , which are real numbers, are related to the angles for the $\vec{J}^\pm$ generator of SU(2) $\times$ SU(2) by $\theta_j^\pm = \theta_j \mp i\beta_j$. So for a boost, the J_+ and J_- generators get multiplied by imaginary angles, which makes the transformation anti-unitary. Thus, none of the representations of the Lorentz group generated this way will be unitary and therefore, there are no finite-dimensional unitary representations of the Lorentz group.

To construct a unitary field theory, we need unitary representations of the Poincaré group, which are infinite dimensional; the corresponding representations of the Lorentz subgroup of the Poincaré group are also infinite dimensional. We will construct an infinite-dimensional representation by having the basis depend on the momentum p^μ . For fixed momentum, say $p^\mu = (m, 0, 0, 0)$ in the massive case, or $p^\mu = (E, 0, 0, E)$ in the massless case, the group reduces to the appropriate little group, SO(3) or ISO(2) respectively. These little groups do have unitary representations. Implementing this procedure for spin 1, we were led uniquely to Lagrangians with kinetic terms of the form $-\frac{1}{4}F_{\mu\nu}^2$, and gauge invariance and charge conservation if $m = 0$. We will now see how to construct Lorentz-invariant Lagrangians that describe unitary theories with spinors.

Lorentz-Invariant Lagrangians

Now, we know that we need infinite-dimensional representations. Therefore, it is natural to introduce spinor-valued fields:

$$\left(\frac{1}{2}, 0\right) : \psi_L(x) = \begin{pmatrix} \psi_1(x) \\ \psi_2(x) \end{pmatrix} \quad (2.17)$$

or

$$\left(0, \frac{1}{2}\right) : \psi_R(x) = \begin{pmatrix} \psi_1(x) \\ \psi_2(x) \end{pmatrix}. \quad (2.18)$$

Similarly to the spin-1 case, we would first like to write down a Lorentz-invariant Lagrangian with the correct number of degrees of freedom, which is two. A first guess would be

$$\mathcal{L} = \psi_R^\dagger \square \psi_R + m^2 \psi_R^\dagger \psi_R. \quad (2.19)$$

Under an infinitesimal Lorentz transformation, (2.14) and (2.15), the second term becomes

$$\begin{aligned} \delta(\psi_R^\dagger \psi_R) &= \frac{1}{2}(-i\theta_j + \beta_j)\psi_R^\dagger \sigma_j \psi_R + \frac{1}{2}\psi_R^\dagger (i\theta_j + \beta_j)\sigma_j \psi_R \\ &= \beta_j \psi_R^\dagger \sigma_j \psi_R \neq 0, \end{aligned} \quad (2.20)$$

which is clearly not Lorentz-invariant. We could also try the term $\psi_L^\dagger \psi_R$. Under an infinitesimal transformation, it gives

$$\begin{aligned}\delta(\psi_L^\dagger \psi_R) &= \frac{1}{2}(-i\theta_j - \beta_j)\psi_L^\dagger \sigma_j \psi_R + \frac{1}{2}\psi_L^\dagger (i\theta_j + \beta_j)\sigma_j \psi_R \\ &= 0,\end{aligned}\tag{2.21}$$

which is great! To get a real Lagrangian we need to add the Hermitian conjugate, which is

$$\mathcal{L}_{\text{Dirac mass}} = m^2(\psi_L^\dagger \psi_R + \psi_R^\dagger \psi_L).\tag{2.22}$$

This term is known as the Dirac mass. However, this is not enough; we still need kinetic terms to get dynamics out of the Lagrangian. Let us look at $\psi_R^\dagger \sigma_i \psi_R$, which transforms as

$$\begin{aligned}\delta(\psi_R^\dagger \sigma_i \psi_R) &= \frac{1}{2}(-i\theta_j + \beta_j)\psi_R^\dagger \sigma_j \sigma_i \psi_R + \frac{1}{2}\psi_R^\dagger (i\theta_j + \beta_j)\sigma_i \sigma_j \psi_R \\ &= \frac{i\theta_j}{2}\psi_R^\dagger (\sigma_i \sigma_j - \sigma_j \sigma_i) \psi_R + \frac{\beta_j}{2}\psi_R^\dagger (\sigma_i \sigma_j + \sigma_j \sigma_i) \psi_R \\ &= \beta_i \psi_R^\dagger \psi_R - \theta_j \varepsilon_{ijk} \psi_R^\dagger \sigma_k \psi_R.\end{aligned}\tag{2.23}$$

Thus, we have found that

$$\delta(\psi_R^\dagger \psi_R, \psi_R^\dagger \sigma_i \psi_R) = (\beta_i \psi_R^\dagger \sigma_j \psi_R, \beta_i \psi_R^\dagger \psi_R - \theta_j \varepsilon_{ijk} \psi_R^\dagger \sigma_k \psi_R)\tag{2.24}$$

transforms exactly like a 4-vector:

$$\delta(V_0, V_i) = (\beta_i V_i, \beta_i V_0 - \varepsilon_{ijk} \theta_j V_k).\tag{2.25}$$

Therefore,

$$\partial_\mu(\psi_R^\dagger \psi_R, \psi_R^\dagger \vec{\sigma} \psi_R)_\mu = \partial_t(\psi_R^\dagger \psi_R) + \partial_j(\psi_R^\dagger \sigma_j \psi_R)\tag{2.26}$$

is Lorentz invariant. However, it is a total derivative that does not contribute to the action. Instead, we can write down

$$\psi_R^\dagger \partial_t \psi_R + \psi_R^\dagger \partial_j \sigma_j \psi_R,\tag{2.27}$$

which is also Lorentz invariant. Similarly, we can write another Lorentz-invariance term with the left-handed Weyl spinor, ψ_L :

$$\psi_L^\dagger \partial_t \psi_L - \psi_L^\dagger \partial_j \sigma_j \psi_L.\tag{2.28}$$

Defining

$$\sigma^\mu \equiv (1, \vec{\sigma}), \quad \bar{\sigma}^\mu \equiv (1, -\vec{\sigma}),\tag{2.29}$$

we can write all the Lorentz-invariant terms we have found as

$$\mathcal{L} = i\psi_R^\dagger \sigma^\mu \partial_\mu \psi_R + i\psi_L^\dagger \bar{\sigma}^\mu \partial_\mu \psi_L - m(\psi_R^\dagger \psi_L + \psi_L^\dagger \psi_R).\tag{2.30}$$

We added a factor of i in the kinetic term to make the Lagrangian Hermitian:

$$(i\psi_R^\dagger \sigma^\mu \partial_\mu \psi_R)^\dagger = -i(\partial_\mu \psi_R^\dagger) \sigma^\mu \psi_R = i\psi_R^\dagger \sigma^\mu \partial_\mu \psi_R, \quad (2.31)$$

where we have used $\sigma_\mu^\dagger = \sigma_\mu$ and integrated by parts.

There is an even shorter-hand way to write this. Let us combine the two spinors into a four-component object known as a Dirac spinor:

$$\psi = \begin{pmatrix} \psi_L \\ \psi_R \end{pmatrix}. \quad (2.32)$$

If we also define

$$\bar{\psi} = (\psi_R^\dagger, \psi_L^\dagger), \quad (2.33)$$

and use the 4×4 matrices

$$\gamma^\mu = \begin{pmatrix} 0 & \sigma^\mu \\ \bar{\sigma}^\mu & 0 \end{pmatrix}, \quad (2.34)$$

known as Dirac matrices or γ -matrices, our Lagrangian becomes

$$\mathcal{L} = \bar{\psi}(i\gamma^\mu \partial_\mu - m)\psi, \quad (2.35)$$

which is the conventional form of the Dirac Lagrangian. The equations of motion that follow are

$$(i\gamma^\mu \partial_\mu - m)\psi = 0, \quad (2.36)$$

which is the Dirac equation.

Dirac Matrices

The Dirac matrices satisfy

$$\{\gamma^\mu, \gamma^\nu\} = 2g^{\mu\nu}. \quad (3.1)$$

Recall, we have used a specific representation to determine the general algebra of the Lorentz group. Here, the algebra of the γ -matrices is more fundamental than any particular representation of them. The γ -matrices generate the Dirac algebra, which is a special case of the Clifford algebra. The particular representation we came across is called the Weyl representation.

Next, for simplicity, we define

$$\sigma^{\mu\nu} \equiv \frac{i}{2}[\gamma^\mu, \gamma^\nu], \quad (3.2)$$

which can be expanded in σ -matrices:

$$\begin{aligned} \sigma^{\mu\nu} &= \frac{i}{2} \left[\begin{pmatrix} \sigma^\mu & \\ \bar{\sigma}^\mu & \end{pmatrix} \begin{pmatrix} \sigma^\nu & \\ \bar{\sigma}^\nu & \end{pmatrix} - \begin{pmatrix} \sigma^\nu & \\ \bar{\sigma}^\nu & \end{pmatrix} \begin{pmatrix} \sigma^\mu & \\ \bar{\sigma}^\mu & \end{pmatrix} \right] \\ &= \frac{i}{2} \begin{pmatrix} \sigma^\mu \bar{\sigma}^\nu - \sigma^\nu \bar{\sigma}^\mu & \\ & \bar{\sigma}^\mu \sigma^\nu - \bar{\sigma}^\nu \sigma^\mu \end{pmatrix} \end{aligned} \quad (3.3)$$

Thus, we find

$$\sigma^{\mu\mu} = 0, \quad (3.4)$$

$$\sigma^{ij} = -\frac{i}{2} \begin{pmatrix} [\sigma^i, \sigma^j] & \\ & [\sigma^i, \sigma^j] \end{pmatrix} = \varepsilon_{ijk} \begin{pmatrix} \sigma_k & \\ & \sigma_k \end{pmatrix}, \quad (3.5)$$

and

$$\sigma^{0i} = -\sigma^{i0} = -i \begin{pmatrix} \sigma_i & \\ & -\sigma_i \end{pmatrix}. \quad (3.6)$$

Recall, in the 4-vector representations, we could embed the generators into an anti-symmetric rank-2 tensor, $V^{\mu\nu}$. Here, we could do the exact same thing. Since the Lorentz generators are $\frac{\sigma_i}{2}$ for the spinor representations, they can be written as

$$S^{\mu\nu} = \frac{1}{2} \sigma^{\mu\nu}. \quad (3.7)$$

More generally, $S^{\mu\nu}$ will satisfy the Lorentz algebra when constructed from any γ -matrices satisfying the Clifford algebra. That is, you can derive from $\{\gamma^\mu, \gamma^\nu\} = 2g^{\mu\nu}$ that

$$[S^{\mu\nu}, S^{\rho\sigma}] = i(g^{\nu\rho} S^{\mu\sigma} - g^{\mu\rho} S^{\nu\sigma} - g^{\nu\sigma} S^{\mu\rho} + g^{\mu\sigma} S^{\nu\rho}). \quad (3.8)$$

It is important to appreciate that the matrices $S_{\mu\nu}$ are different from the matrices $V_{\mu\nu}$ corresponding to the Lorentz generators in the 4-vector representation. In particular, $S_{\mu\nu}$ are complex. So we have found two inequivalent four-dimensional representations. In each case, the group element is determined by six real angles $\theta_{\mu\nu}$ (three rotations and three boosts). The vector or $(\frac{1}{2}, \frac{1}{2})$ representation is irreducible, and has Lorentz group element

$$\Lambda_V = \exp(i\theta_{\mu\nu} V^{\mu\nu}), \quad (3.9)$$

while the Dirac or $(\frac{1}{2}, 0) \oplus (0, \frac{1}{2})$ representation is reducible and has Lorentz group elements

$$\Lambda_s = \exp(i\theta_{\mu\nu} S^{\mu\nu}). \quad (3.10)$$

There are actually a number of Dirac representations, depending on the form of the γ -matrices. We will consider two: the Weyl and Majorana representations.

The Weyl representation has the Lorentz generators

$$S^{ij} = \frac{1}{2} \varepsilon_{ijk} \begin{pmatrix} \sigma_k & \\ & \sigma_k \end{pmatrix}, \quad K^i = S^{0i} = -\frac{i}{2} \begin{pmatrix} \sigma_i & \\ & -\sigma_i \end{pmatrix}, \quad (3.11)$$

which are block diagonal. These are the same generators we used for the $(\frac{1}{2}, 0)$ (the bottom right matrix element) and $(0, \frac{1}{2})$ (the top left matrix element) representations above. This makes it clear that the Dirac representation of the Lorentz group is reducible; it is the direct sum of a left-handed and a right-handed spinor representation.

The Weyl spinors, ψ_L and ψ_R , are in a way more fundamental than Dirac spinors such as ψ because they correspond to irreducible representations of the Lorentz group. But the electron is a Dirac spinor. More importantly, QED is symmetric under $L \leftrightarrow R$. Thus, for QED the γ -matrices make calculations a lot easier than separating out the ψ_L and ψ_R components.

Lorentz Transformation Properties

When dealing with the Dirac matrices, we often suppress the spinor indices. If we make the spinor indices explicit, (3.1) becomes

$$\{\gamma_{\alpha\gamma}^\mu, \gamma_{\gamma\beta}^\nu\} = 2g^{\mu\nu} \delta_{\alpha\beta}. \quad (3.12)$$

Since $\bar{\psi}\gamma^\mu\psi = \psi_L\bar{\sigma}^\mu\psi_L + \psi_R\sigma^\mu\psi_R$ transforms like a 4-vector, we can deduce that

$$\Lambda_s^{-1}\gamma^\mu\Lambda_s = (\Lambda_V)^{\mu\nu}\gamma^\nu, \quad (3.13)$$

where the Λ_s denotes the Lorentz transformation acting on the spinors and Λ_V are the Lorentz transformation acting on the 4-vectors. Making the spinor indices explicit, this becomes

$$(\Lambda_s^{-1})_{\alpha\delta}\gamma_{\delta\gamma}^\mu(\Lambda_s)_{\gamma\beta} = (\Lambda_V)^{\mu\nu}\gamma_{\alpha\beta}^\nu. \quad (3.14)$$

It would be more general to study the Lorentz transformation properties just from the Dirac algebra itself without using any specific representations. First, note that

$$\{\gamma^\mu, \gamma^\nu\} = 2g^{\mu\nu} \implies (\gamma^0)^2 = \mathbb{1}, \quad (\gamma^i)^2 = -\mathbb{1}. \quad (3.15)$$

This implies, that

$$\gamma^{0\dagger} = \gamma^0, \quad \gamma^{i\dagger} = -\gamma^i, \quad (3.16)$$

which says that for any representations, γ^0 is Hermitian and γ^i are anti-Hermitian.

Then,

$$(S^{\mu\nu})^\dagger = \left(\frac{i}{4} [\gamma^\mu, \gamma^\nu] \right)^\dagger = -\frac{i}{4} [\gamma^{\nu\dagger}, \gamma^{\mu\dagger}] = \frac{i}{4} [\gamma^{\mu\dagger}, \gamma^{\nu\dagger}], \quad (3.17)$$

which implies

$$S^{ij\dagger} = S^{ij}, \quad S^{0i\dagger} = -S^{0i}. \quad (3.18)$$

Therefore, we have shown that rotations are Hermitian and boosts are anti-Hermitian in a completely algebraic way, using only the relation $\{\gamma^\mu, \gamma^\nu\} = 2g^{\mu\nu}$. Hence, this must be true for all representations of the Dirac algebra.

Now, using $\{\gamma^\mu, \gamma^\nu\} = 2g^{\mu\nu}$ once again, we find

$$\gamma^0 \gamma^i \gamma^0 = -\gamma^i = \gamma^{i\dagger}, \quad \gamma^0 \gamma^0 \gamma^0 = \gamma^0 = \gamma^{0\dagger}, \quad (3.19)$$

so

$$\gamma^0 \gamma^\mu \gamma^0 = \gamma^{\mu\dagger}. \quad (3.20)$$

Then,

$$\gamma^0 (S^{\mu\nu})^\dagger \gamma^0 = \gamma^0 \frac{i}{4} [\gamma^{\mu\dagger}, \gamma^{\nu\dagger}] \gamma^0 = S^{\mu\nu}, \quad (3.21)$$

and thus

$$(\gamma^0 \Lambda_s \gamma^0)^\dagger = \gamma^0 \exp(-i\theta_{\mu\nu} S^{\mu\nu\dagger}) \gamma^0 = \exp(-i\theta_{\mu\nu} S^{\mu\nu}) = \Lambda_s^{-1}. \quad (3.22)$$

Finally,

$$\psi^\dagger \gamma^0 \psi \rightarrow (\psi^\dagger \Lambda_s^\dagger) \gamma^0 (\Lambda_s \psi) = \psi^\dagger \gamma^0 \Lambda_s^{-1} \Lambda_s \psi = \psi^\dagger \gamma^0 \psi, \quad (3.23)$$

which is Lorentz invariant.

We have just been re-deriving from the Dirac algebra point of view what we found by hand from the Weyl point of view. We have seen that the natural conjugate for ψ out of which real Lorentz-invariant expressions are constructed is not $\psi^\dagger$ but

$$\bar{\psi} \equiv \psi^\dagger \gamma^0. \quad (3.24)$$

The point is that $\bar{\psi}$ transforms according to Λ_s^{-1} . Thus $\bar{\psi}\psi$ is Lorentz invariant. In contrast, $\psi^\dagger\psi$ is not Lorentz invariant, since $\psi^\dagger\psi \rightarrow (\psi^\dagger \Lambda_s^\dagger)(\Lambda_s \psi)$. For this to be invariant, we would need $\Lambda_s^\dagger = \Lambda_s^{-1}$, that is, for the representation of the Lorentz group to be unitary.

But the finite-dimensional spinor representation of the Lorentz group, like the 4-vector representation, is not unitary, because the boost generators are anti-Hermitian. As with vectors, for unitary representations we will need fields $\psi(x)$ that transform in infinite-dimensional representations of the Poincaré group.

We can also construct objects such as

$$\bar{\psi} \gamma_\mu \psi, \quad \bar{\psi} \gamma_\mu \gamma_\nu \psi, \quad \bar{\psi} \partial_\mu \psi; \quad (3.25)$$

all transform like tensors under the Lorentz group. Also

$$\mathcal{L} = \bar{\psi}(i\gamma^\mu \partial_\mu - m)\psi \quad (3.26)$$

is Lorentz invariant. We abbreviate this with

$$\mathcal{L} = \bar{\psi}(i\not{\partial} - m)\psi, \quad (3.27)$$

which is the Dirac Lagrangian.

The Dirac equation follows from this Lagrangian by the equations of motion:

$$(i\not{\partial} - m)\psi = 0. \quad (3.28)$$

To be explicit, this is shorthand for

$$(i\gamma^\mu_{\alpha\beta}\partial_\mu - m\delta_{\alpha\beta})\psi_\beta = 0. \quad (3.29)$$

After multiplying the Dirac equation by $(i\not{\partial} + m)$ we find

$$\begin{aligned} 0 &= (i\not{\partial} + m)(i\not{\partial} - m)\psi \\ &= \left(-\frac{1}{2}\partial_\mu\partial_\nu\{\gamma^\mu, \gamma^\nu\} - \frac{1}{2}\partial_\mu\partial_\nu[\gamma^\mu, \gamma^\nu] - m^2\right)\psi \\ &= -(\partial^2 + m^2)\psi. \end{aligned} \quad (3.30)$$

So ψ satisfies the Klein-Gordon equation:

$$(\square + m^2)\psi = 0. \quad (3.31)$$

In Fourier space, this implies that on-shell spinor momenta satisfy the unique relativistic dispersion relation $p^2 = m^2$, just like scalars. Because spinors also satisfy an equation linear in derivatives, people sometimes say the Dirac equation is the “square root” of the Klein-Gordon equation.

We can integrate the Lagrangian by parts to derive the equations of motion for $\bar{\psi}$:

$$\mathcal{L} = \bar{\psi}i\not{\partial}\psi - m\bar{\psi}\psi = -i(\partial_\mu\bar{\psi})\gamma^\mu\psi - m\bar{\psi}\psi. \quad (3.32)$$

So,

$$-i\partial_\mu\bar{\psi}\gamma^\mu - m\bar{\psi} = 0. \quad (3.33)$$

This γ^μ on the opposite side from ∂_μ is a little annoying, so we often write

$$\bar{\psi}(-i\overleftarrow{\not{\partial}} - m) = 0, \quad (3.34)$$

where the derivative acts to the left. This makes the conjugate equation look more like the original Dirac equation.

Coupling with the Photon Fields

The spinor fields transform exactly the same as for a complex scalar field under a gauge transformation. For example, for a particle with charge $Q = -1$, such as the electrons, we have

$$\psi \rightarrow e^{-i\alpha}\psi. \quad (4.1)$$

Therefore, to couple with the photon fields, we can also replace $\partial_\mu \rightarrow D_\mu$, where $D_\mu = \partial_\mu + ieA_\mu$. Thus, the Lagrangian becomes

$$\mathcal{L} = \bar{\psi}(i\not{\partial} - e\not{A} - m)\psi = \bar{\psi}(i\not{D} - m)\psi, \quad (4.2)$$

and thus the equation of motion becomes

$$(i\not{D} - e\not{A} - m)\psi = (i\not{D} - m)\psi = 0. \quad (4.3)$$

If we compare the Dirac equation to the Klein-Gordon equation of the complex scalar field coupled with the photon field, we will notice something very interesting. The Klein-Gordon equation for the complex scalar field is

$$-(D^2 + m^2)\phi = [(i\partial_\mu - eA_\mu)^2 - m^2]\phi = 0. \quad (4.4)$$

Similarly to before, we multiply the Dirac equation by $i\not{D} - e\not{A} + m$, and thus

$$\begin{aligned} 0 &= (i\not{D} - e\not{A} + m)(i\not{D} - e\not{A} - m)\psi \\ &= (i\gamma^\mu\partial_\mu - e\gamma^\mu A_\mu + m)(i\gamma^\nu\partial_\nu - e\gamma^\nu A_\nu - m)\psi \\ &= [(i\partial_\mu - eA_\mu)(i\partial_\nu - eA_\nu)\gamma^\mu\gamma^\nu - m^2]\psi \\ &= \left(\frac{1}{2}\{i\partial_\mu - eA_\mu, i\partial_\nu - eA_\nu\}\gamma^\mu\gamma^\nu + \frac{1}{2}[i\partial_\mu - eA_\mu, i\partial_\nu - eA_\nu]\gamma^\mu\gamma^\nu - m^2 \right)\psi \\ &= \left(\frac{1}{4}\{i\partial_\mu - eA_\mu, i\partial_\nu - eA_\nu\}\{\gamma^\mu, \gamma^\nu\} + \frac{1}{4}\{i\partial_\mu - eA_\mu, i\partial_\nu - eA_\nu\}[\gamma^\mu, \gamma^\nu] \right. \\ &\quad \left. + \frac{1}{4}[i\partial_\mu - eA_\mu, i\partial_\nu - eA_\nu]\{\gamma^\mu, \gamma^\nu\} + \frac{1}{4}[i\partial_\mu - eA_\mu, i\partial_\nu - eA_\nu][\gamma^\mu, \gamma^\nu] - m^2 \right)\psi \\ &= \left(\frac{1}{4}\{i\partial_\mu - eA_\mu, i\partial_\nu - eA_\nu\}\{\gamma^\mu, \gamma^\nu\} + \frac{1}{4}[i\partial_\mu - eA_\mu, i\partial_\nu - eA_\nu][\gamma^\mu, \gamma^\nu] - m^2 \right)\psi. \end{aligned} \quad (4.5)$$

Now, let us evaluate the commutator

$$\begin{aligned} &= (i\partial_\mu - eA_\mu)(i\partial_\nu - eA_\nu) - (i\partial_\nu - eA_\nu)(i\partial_\mu - eA_\mu) \\ &= -ie(\partial_\mu A_\nu + A_\mu\partial_\nu - \partial_\nu A_\mu - A_\nu\partial_\mu) \\ &= -ie[(\partial_\mu A_\nu) + \cancel{A_\nu\partial_\mu} + \cancel{A_\mu\partial_\nu} - (\partial_\nu A_\mu) - \cancel{A_\mu\partial_\nu} - \cancel{A_\nu\partial_\mu}] \\ &= -ieF_{\mu\nu}. \end{aligned} \quad (4.6)$$

And using the identity $\{\gamma^\mu, \gamma^\nu\} = 2g^{\mu\nu}$, the anti-commutator becomes

$$\{i\partial_\mu - eA_\mu, i\partial_\nu - eA_\nu\}\{\gamma^\mu, \gamma^\nu\} = 4(i\partial_\mu - eA_\mu)^2. \quad (4.7)$$

Thus, the Dirac equation becomes

$$\begin{aligned} 0 &= \left((i\partial_\mu - eA_\mu)^2 - \frac{i}{4}eF_{\mu\nu}[\gamma^\mu, \gamma^\nu] - m^2 \right) \psi \\ &= \left((i\partial_\mu - eA_\mu)^2 - \frac{e}{2}F_{\mu\nu}\sigma^{\mu\nu} - m^2 \right) \psi, \end{aligned} \quad (4.8)$$

where we have used the definition $\sigma^{\mu\nu} = [\gamma^\mu, \gamma^\nu]$. Comparing (4.8) and (4.4), we see that there is an extra $\frac{e}{2}F_{\mu\nu}\sigma^{\mu\nu}$ term for the spinor QED. This can be written as

$$\not{D}^2 = D_\mu^2 + \frac{e}{2}F_{\mu\nu}\sigma^{\mu\nu}, \quad (4.9)$$

which tells us the difference between the covariant derivative when acting on the scalar field and the spinor field. So, what is this extra term? In the Weyl representation, they have the form

$$\begin{aligned} \frac{e}{2}F_{\mu\nu}\sigma^{\mu\nu} &= \frac{e}{2}(F^{0i}\sigma^{0i} - F^{i0}\sigma^{i0} + F^{ij}\sigma^{ij}) \\ &= e \left[-iE_i \begin{pmatrix} \sigma_i & 0 \\ 0 & -\sigma_i \end{pmatrix} - \frac{1}{2}\varepsilon_{ijl}B_l\varepsilon_{ijk} \begin{pmatrix} \sigma_k & 0 \\ 0 & \sigma_k \end{pmatrix} \right] \\ &= e \left[-iE_i \begin{pmatrix} \sigma_i & 0 \\ 0 & -\sigma_i \end{pmatrix} - \delta_{lk}B_l \begin{pmatrix} \sigma_k & 0 \\ 0 & \sigma_k \end{pmatrix} \right] \\ &= -e \begin{pmatrix} (\vec{B} + i\vec{E}) \cdot \vec{\sigma} & 0 \\ 0 & (\vec{B} - i\vec{E}) \cdot \vec{\sigma} \end{pmatrix}. \end{aligned} \quad (4.10)$$

Thus, the Dirac equation can be written explicitly as

$$\left\{ (\partial_\mu + ieA_\mu)^2 + m^2 - e \begin{pmatrix} (\vec{B} + i\vec{E}) \cdot \vec{\sigma} & 0 \\ 0 & (\vec{B} - i\vec{E}) \cdot \vec{\sigma} \end{pmatrix} \right\} \psi = 0. \quad (4.11)$$

Finally, note that the Noether current associated with the global symmetry $\psi \rightarrow e^{-i\alpha}\psi$ is

$$J_\mu = \bar{\psi}\gamma_\mu\psi. \quad (4.12)$$

This, like any Noether current, is conserved on the equations of motion even if we set $A_\mu = 0$. The 0 component of this current gives the charge density

$$J_0 = \psi^\dagger\psi = \psi_L^\dagger\psi_L + \psi_R^\dagger\psi_R. \quad (4.13)$$

We originally hoped this would be Lorentz invariant, which it is not. Now we see that it transforms as the 0 component of a conserved current. We can interpret this as the probability density for a fermion. The conserved charge $Q = \int d^3x J_0$ is electron number, which is the number of electrons minus the number of positrons. The spatial components of J_μ denote electron number flow. The electron number current J_μ is related to the charge current eJ_μ , which couples to A_μ , by a factor of the electric charge e .

The Magnetic Dipole Moment

The Dirac equation predicted that there is an interaction between the magnetic field $\vec{B}$ and the electron spin $\vec{S}$, which is known as the magnetic dipole moment of the electron. Before the Dirac equation, people did not understand spin and were particularly confused about the origin of the electron's magnetic dipole moment. To understand it, let us first take the non-relativistic limit. In quantum mechanics, we need to work with the Hamiltonian instead of the Lagrangian. Of course, one can find the Dirac Hamiltonian in the regular way. However, it is easier to express the Dirac equation in the form $i\partial_t\psi = H_D\psi$, so that we have the Dirac Hamiltonian automatically. The Dirac equation can be written as

$$(i\gamma^0\partial_t + i\gamma^i\partial_i - e\cancel{A} - m)\psi = 0, \quad (4.14)$$

which implies

$$i\partial_t\psi = [\gamma^0\gamma^i(-i\partial_i - eA_i) + eA_0 + m\gamma^0]\psi = H_D\psi. \quad (4.15)$$

Replace $-i\partial_i$ by p_i , the Dirac Hamiltonian becomes

$$\begin{aligned} H_D &= \gamma^0\gamma^i(p_i - eA_i) + eA_0 + m\gamma^0 \\ &= \gamma^0\gamma^i\Pi_i + eA_0 + m\gamma^0, \end{aligned} \quad (4.16)$$

where $\Pi_i \equiv p_i - eA_i$ is known as the kinetic momentum.

Now we compute $(H_D - eA_0)^2$ in the Weyl representations:

$$\begin{aligned} (H_D - eA_0)^2 &= (\gamma^0\gamma^i\Pi_i + m\gamma^0)(\gamma^0\gamma^j\Pi_j + m\gamma^0) \\ &= \gamma^0\gamma^i\gamma^0\gamma^j\Pi_i\Pi_j + \gamma^0\gamma^i\gamma^0m\Pi_i + \gamma^0\gamma^0\gamma^j m\Pi_j + m^2 \\ &= -\gamma^i\gamma^j\Pi_i\Pi_j - \cancel{\gamma^i m\Pi_i} + \cancel{\gamma^j m\Pi_j} + m^2 \\ &= -\begin{pmatrix} -\sigma^i\sigma^j & \\ & -\sigma^i\sigma^j \end{pmatrix}\Pi_i\Pi_j + m^2 \\ &= [(\vec{\sigma} \cdot \vec{\Pi})^2 + m^2]\mathbb{1}. \end{aligned} \quad (4.17)$$

Putting back $\hbar$ and c , the above becomes

$$(H_D - eA_0)^2 = [(\vec{\sigma} \cdot \vec{\Pi})^2 c^2 + m^2 c^4]\mathbb{1} \quad (4.18)$$

Then, subtract off the zero-point energy mc^2 from H_D , and expand to order c^0 , we have

$$\begin{aligned} H &= H_D - mc^2 = \sqrt{(H_D - eA_0)^2 + ecA_0 - mc^2} \\ &= \left[\sqrt{(\vec{\sigma} \cdot \vec{\Pi})^2 c^2 + m^2 c^4 + ecA_0 - mc^2} \right] \mathbb{1}. \end{aligned} \quad (4.19)$$

Since

$$\begin{aligned} \sqrt{(\vec{\sigma} \cdot \vec{\Pi})^2 c^2 + m^2 c^4} &= mc^2 \left[1 + \frac{1}{2m^2 c^2} (\vec{\sigma} \cdot \vec{\Pi})^2 + \dots \right] \\ &= mc^2 + \frac{1}{2m} (\vec{\sigma} \cdot \vec{\Pi})^2 + \mathcal{O}\left(\frac{1}{c^2}\right), \end{aligned} \quad (4.20)$$

we find

$$H = H_D - mc^2 = \frac{1}{2m}(\vec{\sigma} \cdot \vec{\Pi})^2 + ecA_0 + \mathcal{O}\left(\frac{1}{c^2}\right), \quad (4.21)$$

where we have suppressed the identity matrix $\mathbb{1}$. To see the coupling strength between the magnetic field and the spin, we need to further simplify the above relation. Note,

$$(\vec{\sigma} \cdot \vec{\Pi})^2 = \sigma^i \sigma^j \Pi_i \Pi_j = \frac{1}{2}(\{\sigma^i, \sigma^j\} + [\sigma^i, \sigma^j])\Pi_i \Pi_j = (\delta^{ij} + i\varepsilon^{ijk}\sigma_k)\Pi_i \Pi_j. \quad (4.22)$$

Then, putting back $p_i \rightarrow -i\partial_i$, we find

$$\begin{aligned} \varepsilon^{ijk}\sigma_k \Pi_i \Pi_j &= \varepsilon^{ijk}\sigma_k (i\partial_i + eA_i)(i\partial_j + eA_j) \\ &= \varepsilon^{ijk}\sigma_k [-\partial_i \partial_j + ie\partial_i A_j + ieA_i \partial_j + e^2 A_i A_j] \\ &= -\sigma_k (\partial_i \partial_j - \partial_j \partial_i - \cancel{A_i A_j + A_j A_i}) + ie\varepsilon^{ijk}\sigma_k [A_j \partial_i + (\partial_i A_j) + A_i \partial_j] \\ &= ie\varepsilon^{ijk}\sigma_k (\partial_i A_j) \\ &= ieB^k \sigma_k. \end{aligned} \quad (4.23)$$

Thus, the Hamiltonian becomes (putting back the $\hbar$ in the above derivation)

$$H = \frac{1}{2m}(p_i - eA_i)^2 - \frac{e}{m}\vec{B} \cdot \vec{S} + ecA_0 + \mathcal{O}\left(\frac{1}{c^2}\right). \quad (4.24)$$

From this, we can see that the spin magnetic dipole moment of an electron is $-\frac{e}{m}$ to order of c^0 .

Meanwhile, the spin electric dipole moment is simply 0, because we do not see any $\vec{E} \cdot \vec{S}$ coupling above.

Now, if we choose the magnetic field to be constant in the z -direction, then $\vec{B} = B_z \hat{k}$. Thus, we can set $\vec{A} = -B_z y/2\hat{i} + B_z x/2\hat{j}$. The Hamiltonian becomes

$$\begin{aligned} H &= \frac{\vec{p}^2}{2m} + \frac{e^2}{2m}\vec{A}^2 - \frac{e}{2m}\left(-\frac{B_z y p_x}{2} + \frac{B_z x p_y}{2}\right) - \frac{e}{m}B_z S_z + ecA_0 + \mathcal{O}\left(\frac{1}{c^2}\right) \\ &= \frac{\vec{p}^2}{2m} + \frac{e^2}{2m}\vec{A}^2 - \frac{e}{2m}(L_z + 2S_z)B_z + ecA_0 + \mathcal{O}\left(\frac{1}{c^2}\right). \end{aligned} \quad (4.25)$$

Note there is also a coupling term between the magnetic field $\vec{B}$ and the angular momentum $\vec{L}$ with the coupling strength $\mu_B = -\frac{e}{2m}$. Now, we define the gyromagnetic ratio, g_e , to be the relative size of the $\vec{S} \cdot \vec{B}$ and $\vec{L} \cdot \vec{B}$ interactions. Thus, we have just found out that $g_e = 2$ to the 0th order of c , since the entire coupling to the magnetic field is given by

$$\vec{\mu} \equiv \mu_B(\vec{L} + g_e \vec{S}) \cdot \vec{B} = -\frac{e}{2m}(\vec{L} + 2\vec{S}) \cdot \vec{B}. \quad (4.26)$$

The Meaning of spin- $\frac{1}{2}$

To understand spin- $\frac{1}{2}$ particles, let us begin by looking at what happens when we rotate them by an angle θ in the z plane. For any representation, such a rotation is given by

$$\Lambda(\theta_z) = \exp(i\theta_z J_z), \quad (5.1)$$

with J_z the generator in an appropriate representation. The easiest way to exponentiate a matrix is to first diagonalize it with a unitary transformation, then exponentiate the eigenvalues, then transform back. This unitary transformation is like choosing a (possibly complex) direction. If we are only ever rotating around one axis, we can simply use the diagonal basis for the exponentiation.

First, for the vector representation,

$$J_3 = V_{12} = i \begin{pmatrix} 0 & & & \\ & 0 & -1 & \\ & 1 & 0 & \\ & & & 0 \end{pmatrix} = U^{-1} \begin{pmatrix} 0 & & & \\ & -1 & & \\ & & 1 & \\ & & & 0 \end{pmatrix} U. \quad (5.2)$$

Note that the eigenvalues of J_3 are $-1, 0, 1$ and 0 , which is what one expects from the $(\frac{1}{2}, \frac{1}{2})$ representation of the Lorentz group describing spins 1 and 0. So,

$$\Lambda_V(\theta_z) = \exp(i\theta_z V_{12}) = U^{-1} \begin{pmatrix} 1 & & & \\ & \exp(-i\theta_z) & & \\ & & \exp(i\theta_z) & \\ & & & 1 \end{pmatrix} U \quad (5.3)$$

and

$$\Lambda_V(2\pi) = 1. \quad (5.4)$$

That is, we rotate 360 degrees and we are back to where we started.

For the spinor representation

$$\Lambda_s(\theta_z) = \exp(i\theta_z S_{12}) \quad (5.5)$$

the 12 rotation is already diagonal:

$$S_{12} = \begin{pmatrix} \frac{1}{2} & & & \\ & -\frac{1}{2} & & \\ & & \frac{1}{2} & \\ & & & -\frac{1}{2} \end{pmatrix}. \quad (5.6)$$

Here the eigenvalues are $-\frac{1}{2}, \frac{1}{2}, -\frac{1}{2}$ and $\frac{1}{2}$, as one expects for the $(\frac{1}{2}, 0) \oplus (0, \frac{1}{2})$ representation of the Lorentz group. So,

$$\Lambda_s(\theta_z) = \exp(i\theta_z S_{12}) = \begin{pmatrix} \exp(\frac{i}{2}\theta_z) & & & \\ & \exp(-\frac{i}{2}\theta_z) & & \\ & & \exp(\frac{i}{2}\theta_z) & \\ & & & \exp(-\frac{i}{2}\theta_z) \end{pmatrix} \quad (5.7)$$

and

$$\Lambda_s(2\pi) = \begin{pmatrix} -1 & & & \\ & -1 & & \\ & & -1 & \\ & & & -1 \end{pmatrix} = -\mathbb{1}. \quad (5.8)$$

Thus, a 2π rotation does not bring us back where we started! If we rotate by 4π it would, but with a 2π rotation we pick up a -1 .

By the way, this odd factor of -1 is the origin of the connection between spin and statistics. As a quick way to see the connection, consider a state containing two identical fermions localized on opposite sides of the origin in the $\hat{x}$ direction. Let their spins both point in the $+\hat{z}$ direction. So the state is $|\psi_{12}\rangle = |\psi_1(\vec{x})\psi_2(-\vec{x})\rangle$. Now rotate the state around the z axis by π . Such a rotation interchanges the two particles, and does not affect the spins. It also induces a factor of $\Lambda_s(\pi) = i$ for each spinor. Thus, $|\psi_{12}\rangle \rightarrow -|\psi_{21}\rangle$ since the particles are identical, $|\psi_{12}\rangle = |\psi_{21}\rangle$. Thus, the wavefunction picks up a -1 when the particles are interchanged. That is, the spinors are fermions.

Projective Representation

How can something go back to minus itself under a 2π rotation? This is not something that can happen in the Lorentz group. By definition, all representations of the Lorentz group map 2π rotations to the identity element of the group: $r[2\pi] = 1$. And, by definition, the identity group element sends objects to themselves. The problem is that by exponentiating elements of the Lie algebra for the group we generated a different group, $SL(2, \mathbb{C})$, which is the universal cover of the Lorentz group, not the Lorentz group itself. So, technically, spinors transform as representations of $SL(2, \mathbb{C})$. Why is this OK?

Recall that the Lorentz group is defined as the group preserving the Minkowski metric $\Lambda^T g \Lambda = g$. Observables, in particular the S -matrix, should be invariant under this symmetry. In quantum mechanics, we learned that states are identified with rays, so that $|\psi\rangle$ and $\lambda|\psi\rangle$ for any complex number λ are the same state. In field theory, we have carefully normalized our fields (and we will carefully renormalize them), so we do not want that norm to change in different frames. However, we can still have the fields change by a phase without upsetting their norms. Thus, for physical purposes what we are looking for is not exactly representations of the Lorentz group, but **projective representations** of the Lorentz group, in which group elements can change the phase of a state. Projective representations can have

$$r[g_1]r[g_2] = e^{i\phi(g_1, g_2)} r[g_1 g_2], \quad (5.9)$$

which is a generalization of the normal requirement that $r[g_1 g_2] = r[g_1]r[g_2]$ for a group representation. The projective representations of $O(1, 3)$ are the same as the representations of $SL(2, \mathbb{C})$, which include the spinors.

Using objects that have properties that are not directly observable is not new. For example, in quantum mechanics, we learned that wavefunctions are complex. There are plenty of implications of the complexity, but you do not actually measure complex things. In the same way, although we only measure Lorentz-invariant things (matrix elements), the most general theory can have objects, spinors, that are a little bit more complicated than the Lorentz group alone would naively suggest. Although spinors transform in representations of $SL(2, \mathbb{C})$, the Poincaré group is still the symmetry group of observables.

The existence of objects, spinors, that transform as $\psi \rightarrow -\psi$ under $\theta = 2\pi$ rotations is closely related to an interesting fact about the 3D rotation group that you might not be aware of: it is not simply connected. In a group that is not simply connected, there are closed paths through the group that are not contractible, that is, they cannot be smoothly deformed to a point. For example, the group $\text{SO}(2)$ of 2D rotations is specified by angles θ . Let us describe our path by a number t , with $0 \leq t < 1$. Then the path $\theta(t) = 2\pi t$ is not smoothly deformable to $\theta(t) = 0$. That means there is not a smooth function $\theta(t, u)$ for $0 \leq u \leq 1$, so that $\theta(t, 0) = \theta(t)$ and $\theta(t, 1) = 0$. In fact, none of the paths $\theta(t) = 2\pi nt$ for an integer n can be deformed into each other. We say the fundamental group for $\text{SO}(2)$ is $\mathbb{Z}$.

The group $\text{SO}(3)$ is not simply connected either. To see that, define rotations around the z axis by an angle θ_z , and consider the path $\theta_z(t) = 2\pi t$ corresponding to the group elements

$$R(t) = \begin{pmatrix} \cos 2\pi t & \sin 2\pi t & 0 \\ -\sin 2\pi t & \cos 2\pi t & 0 \\ 0 & 0 & 1 \end{pmatrix}. \quad (5.10)$$

This path cannot be smoothly deformed to the identity. Try it! Try to find $R(t, u)$ so that $R(t, 0) = R(t)$ and $R(t, 1) = 1$.

Any rotation in $\text{SO}(3)$ can be specified by an axis $\vec{v}$ and an angle $-\pi \leq \theta_v < \pi$. If we think of the axis as a point on the surface of a ball of radius $r = \pi$, then the rotation can be specified by a point in the ball, with the distance from the origin being the angle θ_v . Thus, a path through the group is a path through the ball. The identity group element is the centre of the ball. There is one catch, however: rotations about an axis by an angle θ_v are the same as rotations about an axis pointing in the opposite direction by the angle $-\theta_v$. Thus, we have to identify antipodal points on the sphere as the same group element in $\text{SO}(3)$. In this sense, $\text{SO}(3)$ is a real projective space $B^3/\mathbb{Z}_2 = \mathbb{RP}^3$. The full ball is $\text{SU}(2)$, which is the universal cover of $\text{SO}(3)$, and $\mathbb{Z}_2$ is the fundamental group of $\text{SO}(3)$ (for the full Lorentz group, the cover is $\text{SL}(2, \mathbb{C})$) and the fundamental group is still $\mathbb{Z}_2$.

Paths from 1 to 1 in $\text{SO}(3)$ go from the centre of the ball back to the centre. Remember that the antipodal points on the sphere are π and $-\pi$ rotations around the same axis, so they are the same group element.

You can actually see these non-contractible paths without too much difficulty by just holding something (like a glass of water) in your hand with your arm outstretched and rotating your arm 360 degrees in a plane parallel to your body. Then your arm (the path) will be twisted, but the object in your hand will have mapped back to itself. You can untangle your arm (the path) with another 360 degree rotation, in this case in a plane parallel to the ground, which gives another $\mathbb{Z}_2$ undoing the twist. If you are careful, you will not even spill the water. Spinors maintain an imprint of how they have been rotated, which shows up as a minus sign after a 2π rotation, much like your arm would if it were an internal degree of freedom of the glass of water. This demonstration is sometimes called Dirac's belt trick, Feynman's plate trick, the Balinese cup trick or the quaternionic handshake.

Majorana and Weyl Fermions

In QED, we only need the electrons, which are efficiently described in the reducible Dirac representation $(0, \frac{1}{2}) \oplus (\frac{1}{2}, 0)$ of the Lorentz group. In other theories, such as the Standard Model or supersym-

metric theories, spinors that are not Dirac spinors are important. In this section, we will discuss other spinors that are not in the Dirac representation.

Majorana Masses

There is another Lorentz-invariant quantity we can construct using ψ_L and ψ_R . The bilinear quantity:

$$\mathcal{L}_{\text{Maj}} = \psi_R^T \sigma_2 \psi_R, \quad (6.1)$$

which is known as the Majorana mass.

To see the Lorentz invariance, recall that for the Pauli matrices σ_1 and σ_3 are real, and σ_2 is imaginary:

$$\sigma_1 = \begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix}, \quad \sigma_2 = \begin{pmatrix} 0 & -i \\ i & 0 \end{pmatrix}, \quad \sigma_3 = \begin{pmatrix} 1 & 0 \\ 0 & -1 \end{pmatrix}. \quad (6.2)$$

So,

$$\sigma_1^* = \sigma_1, \quad \sigma_2^* = -\sigma_2, \quad \sigma_3^* = \sigma_3, \quad (6.3)$$

$$\sigma_1^T = \sigma_1, \quad \sigma_2^T = -\sigma_2, \quad \sigma_3^T = \sigma_3. \quad (6.4)$$

This implies $\sigma_1^T \sigma_2 = \sigma_1 \sigma_2 = -\sigma_2 \sigma_1$, $\sigma_3^T \sigma_2 = \sigma_3 \sigma_2 = -\sigma_2 \sigma_3$ and $\sigma_2^T \sigma_2 = -\sigma_2 \sigma_2$. That is,

$$\sigma_j^T \sigma_2 = -\sigma_2 \sigma_j \quad (6.5)$$

and so

$$\delta(\psi_R^T \sigma_2) = \frac{1}{2}(i\theta_j + \beta_j) \psi_R^T \sigma_j^T \sigma_2 = \frac{1}{2}(-i\theta_j - \beta_j)(\psi_R^T \sigma_2) \sigma_j. \quad (6.6)$$

Combining this with the transformation of ψ_R we see that $\mathcal{L}_{\text{Maj}}$ is Lorentz invariant.

Since $\sigma_2 = \begin{pmatrix} 0 & -i \\ i & 0 \end{pmatrix}$ the Majorana mass can be expanded out to

$$\psi_R^T \sigma_2 \psi_R = (\psi_1, \psi_2) \begin{pmatrix} 0 & -i \\ i & 0 \end{pmatrix} \begin{pmatrix} \psi_1 \\ \psi_2 \end{pmatrix} = -i(\psi_1 \psi_2 - \psi_2 \psi_1). \quad (6.7)$$

Thus, we have shown that $\psi_1 \psi_2 - \psi_2 \psi_1$ is Lorentz invariant. We often write this as

$$\psi_1 \psi_2 - \psi_2 \psi_1 = \psi^\alpha \psi^\beta \epsilon_{\alpha\beta}, \quad \epsilon_{\alpha\beta} = \begin{pmatrix} 0 & 1 \\ -1 & 0 \end{pmatrix}, \quad (6.8)$$

which avoids picking a σ_2 .

There is only one problem: if the fermion components commute, $\psi_1 \psi_2 - \psi_2 \psi_1 = 0$! For Majorana masses to be non-trivial, fermion components cannot be regular numbers; they must be anticommuting numbers. Such things are called Grassmann numbers and satisfy a Grassmann algebra.

Notation for Weyl Spinors

Although, we only need Dirac spinors in QED, the Weyl spinor is still important, since it is the irreducible representation of the Lorentz group. It will be helpful to have a concise notation for constructing products and contractions of Weyl spinors only.

Let us write ψ for left-handed spinors and $\tilde{\psi}$ for right-handed spinors, and we will index the two components of the left-handed Weyl spinor with Greek indices, i.e. ψ_α , and the two components of the right-handed Weyl spinors with dotted Greek indices, i.e. $\psi_{\dot{\alpha}}$. In addition, we will label the left-handed spinor with upper indices. Thus, a Dirac spinor can be written as

$$\begin{pmatrix} \psi^\alpha \\ \psi_{\dot{\beta}} \end{pmatrix}. \quad (6.9)$$

The Majorana mass has the form

$$\mathcal{L}_{\text{Maj}} = \tilde{\psi}^T \sigma_2 \tilde{\psi} = -i(\psi_1 \psi_2 - \psi_2 \psi_1) = i \tilde{\psi}_{\dot{\alpha}} \varepsilon^{\dot{\alpha}\dot{\beta}} \tilde{\psi}_{\dot{\beta}}, \quad (6.10)$$

where we have just defined $\varepsilon^{\alpha\beta} = -i\sigma_2^{\alpha\beta}$. It is a totally antisymmetric 2×2 tensor

$$\varepsilon^{\alpha\beta} = -\varepsilon_{\alpha\beta} = \begin{pmatrix} 0 & 1 \\ -1 & 0 \end{pmatrix}. \quad (6.11)$$

That is, $\varepsilon^{12} = \varepsilon_{21} = 1$, which leads to $\varepsilon_{\alpha\beta} \varepsilon^{\beta\gamma} = \delta_\alpha^\gamma$. The ε tensor serves the function for Weyl spinors that $g_{\mu\nu}$ does for tensors - we can always contract spinors into Lorentz-invariant combinations with the ε tensor. However, we have to be careful raising and lowering indices, since

$$\psi^\alpha \chi_\alpha = \varepsilon^{\alpha\beta} \varepsilon_{\alpha\gamma} \psi_\beta \chi^\gamma = -\varepsilon^{\beta\alpha} \varepsilon_{\alpha\gamma} \psi_\beta \chi^\gamma = -\delta_\gamma^\beta \psi_\beta \chi^\gamma = -\psi_\beta \chi^\beta. \quad (6.12)$$

While it seems that the index position makes things messy, it actually makes things easier, since spinors anticommute. We can define the inner product between two left-handed spinors as

$$\psi \chi \equiv \psi_\alpha \chi^\alpha = \psi_\alpha \varepsilon^{\alpha\beta} \chi_\beta = -\chi_\beta \varepsilon^{\alpha\beta} \psi_\alpha = \chi_\beta \varepsilon^{\beta\alpha} \psi_\alpha = \chi_\alpha \psi^\alpha = \chi \psi, \quad (6.13)$$

so that the product is symmetric. Note that $\psi\psi \neq 0$ even though $\psi^\alpha \psi_\alpha = 0$. For right-handed spinors, we define

$$\tilde{\psi} \tilde{\chi} = \tilde{\psi}^\alpha \tilde{\chi}_\alpha = -\tilde{\chi}_\alpha \tilde{\psi}^\alpha = \tilde{\chi}^\alpha \tilde{\psi}_\alpha = \tilde{\chi} \tilde{\psi}, \quad (6.14)$$

which is also symmetric.

For Weyl spinors, the σ -matrices $\sigma^\mu = (1, \vec{\sigma})$ and $\bar{\sigma}^\mu = (1, -\vec{\sigma})$ replace the Dirac γ -matrices. Recall that the kinetic term for a Dirac spinor $\psi = (\psi, \tilde{\chi})$ is

$$\mathcal{L}_{\text{kin}} = i\tilde{\psi} \not{\partial} \psi = i\psi^\dagger \bar{\sigma}^\mu \partial_\mu \psi + i\tilde{\chi}^\dagger \sigma^\mu \partial_\mu \tilde{\chi}. \quad (6.15)$$

Each of these two terms is separately Lorentz invariant. With spinor indices, $\sigma^\mu = \sigma_{\alpha\dot{\alpha}}^\mu$, the contractions are

$$\tilde{\chi}^\dagger \sigma^\mu \tilde{\chi} = (\tilde{\chi}^\dagger)^{\dot{\alpha}} \sigma_{\dot{\alpha}\alpha}^\mu \tilde{\chi}^\alpha = \chi^\alpha \sigma_{\alpha\dot{\alpha}}^\mu \tilde{\chi}^{\dot{\alpha}}, \quad (6.16)$$

where we have defined a left-handed spinor $\chi \equiv \tilde{\chi}^\dagger$ so that we can drop the $\dagger$. You can think of χ as the particle and $\tilde{\chi}$ as the antiparticle for the same Weyl spinor. Similarly,

$$\psi^\dagger \bar{\sigma}^\mu \psi = (\psi^\dagger)^{\dot{\alpha}} \bar{\sigma}_{\dot{\alpha}\alpha}^\mu \psi^\alpha = \tilde{\psi}^\alpha \bar{\sigma}_{\alpha\dot{\alpha}}^\mu \psi^{\dot{\alpha}} \quad (6.17)$$

with $\tilde{\psi} \equiv \psi^\dagger$.

Two very useful relations are

$$g_{\mu\nu} \sigma_{\alpha\dot{\alpha}}^\mu \sigma_{\beta\dot{\beta}}^\nu = 2\epsilon_{\alpha\beta} \epsilon_{\dot{\alpha}\dot{\beta}} \quad (6.18)$$

and

$$\epsilon_{\alpha\beta} \epsilon_{\dot{\alpha}\dot{\beta}} \sigma^{\mu\beta\dot{\beta}} = \bar{\sigma}_{\dot{\alpha}\alpha}^\mu. \quad (6.19)$$

Spin, Helicity and Chirality

In a relativistic theory, unlike quantum mechanics, spin is a confusing subject. There are three quantities related to spin: spin, chirality and helicity. In this section, we will define and make sense of each of them.

Recall, the Dirac equation can be written in the form

$$\left((i\partial_\mu - eA_\mu)^2 - m^2 - \frac{e}{2} \sigma^{\mu\nu} F_{\mu\nu} \right) \psi = 0, \quad (7.1)$$

and for the conjugate field $\bar{\psi} = \psi^\dagger \gamma^0$

$$\bar{\psi} \left((i\overleftarrow{\partial}_\mu + eA_\mu)^2 - m^2 + \frac{e}{2} \sigma^{\mu\nu} F_{\mu\nu} \right) = 0. \quad (7.2)$$

This means that ψ and $\bar{\psi}$ are a particle-antiparticle pair. We normally call ψ an electron and $\bar{\psi}$ a positron.

When we constructed the Dirac representation, we saw that it was the direct sum of two irreducible representations of the Lorentz group: $(\frac{1}{2}, 0) \oplus (0, \frac{1}{2})$. Now we see that it describes two physically distinguishable particles: the electron and the positron. Irreducible unitary spin- $\frac{1}{2}$ representations of the Poincaré group, Weyl spinors, have two degrees of freedom. Dirac spinors have four. These are two spin states for the electron and two spin states for the positron. For charged spinors, there is no other way. Uncharged spinors can be their own antiparticles if they are Majorana spinors, which we will discuss later.

To understand the degrees of freedom within a four-component Dirac spinor, first recall that in the Weyl basis the γ -matrices have the form

$$\gamma_\mu = \begin{pmatrix} 0 & \sigma_\mu \\ \bar{\sigma}_\mu & 0 \end{pmatrix}, \quad (7.3)$$

and the Lorentz generators $S^{\mu\nu} = \frac{i}{4}[\gamma^\mu, \gamma^\nu]$ are block diagonal. Under an infinitesimal Lorentz transformation,

$$\psi \rightarrow \psi + \frac{1}{2} \begin{pmatrix} (i\theta_i - \beta_i)\sigma_i & \\ & (i\theta_i + \beta_i)\sigma_i \end{pmatrix} \psi, \quad (7.4)$$

In this basis, a Dirac spinor is a doublet of a left- and a right-handed Weyl spinor:

$$\psi = \begin{pmatrix} \psi_L \\ \psi_R \end{pmatrix}. \quad (7.5)$$

Here left-handed and right-handed refer to the $(\frac{1}{2}, 0)$ or $(0, \frac{1}{2})$ representations of the Lorentz group. The handedness of a spinor is also known as its chirality.

It is helpful to be able to project out the left- or right-handed Weyl spinors from a Dirac spinor. We can do that with the γ_5 -matrix:

$$\gamma^5 \equiv i\gamma^0\gamma^1\gamma^2\gamma^3. \quad (7.6)$$

In the Weyl representation

$$\gamma^5 = \begin{pmatrix} -\mathbb{1} & 0 \\ 0 & \mathbb{1} \end{pmatrix}, \quad (7.7)$$

so left- and right-handed spinors are eigenstates of γ_5 with eigenvalues ± 1 . We can also define projection operators,

$$P_R = \frac{1 + \gamma^5}{2} = \begin{pmatrix} 0 & 0 \\ 0 & \mathbb{1} \end{pmatrix}, \quad P_L = \frac{1 - \gamma^5}{2} = \begin{pmatrix} \mathbb{1} & 0 \\ 0 & 0 \end{pmatrix}, \quad (7.8)$$

which satisfy $P_R^2 = P_R$ and $P_L^2 = P_L$ and

$$P_R \begin{pmatrix} \psi_L \\ \psi_R \end{pmatrix} = \begin{pmatrix} 0 \\ \psi_R \end{pmatrix}, \quad P_L \begin{pmatrix} \psi_L \\ \psi_R \end{pmatrix} = \begin{pmatrix} \psi_L \\ 0 \end{pmatrix}. \quad (7.9)$$

Writing projectors as $\frac{1+\gamma^5}{2}$ is basis independent.

It is easy to check that $(\gamma^5)^2 = \mathbb{1}$ and $\{\gamma^5, \gamma^\mu\} = 0$. Thus γ^5 is like another γ -matrix, which is why we call it γ_5 . This lets us formally extend the Clifford algebra to five generators, $\gamma^M = (\gamma^0, \gamma^1, \gamma^2, \gamma^3, i\gamma^5)$ so that $\{\gamma^M, \gamma^N\} = 2g^{MN}$ with $g^{MN} = \text{diag}(1, -1, -1, -1, -1)$.

To understand the degrees of freedom in the spinor, let us focus on the free theory. In the Weyl basis, the Dirac equation is

$$\begin{pmatrix} -m & i\sigma^\mu \partial_\mu \\ i\bar{\sigma}^\mu \partial_\mu & -m \end{pmatrix} \begin{pmatrix} \psi_L \\ \psi_R \end{pmatrix} = 0. \quad (7.10)$$

In Fourier space, this implies

$$\sigma^\mu p_\mu \psi_R = (E - \vec{\sigma} \cdot \vec{p}) \psi_R = m \psi_L, \quad (7.11)$$

$$\bar{\sigma}^\mu p_\mu \psi_L = (E + \vec{\sigma} \cdot \vec{p}) \psi_L = m \psi_R. \quad (7.12)$$

So the mass mixes the left- and right-handed states.

In the absence of a mass, left- and right-handed states are eigenstates of the operator $\hat{h} = \frac{\vec{\sigma} \cdot \vec{p}}{|\vec{p}|}$ with opposite eigenvalue, since $E = |\vec{p}|$ for massless particles. This operator projects the spin on the momentum direction. Spin projected on the direction of motion is called the helicity, so the left- and right-handed states have opposite helicity in the massless theory.

When there is a mass, the left- and right-handed fields mix due to the equations of motion. However, since momentum and spin are good quantum numbers in the free theory, even with a mass, helicity is conserved as well. Therefore, helicity can still be a useful concept for the massive theory. The distinction is that, when there is a mass, helicity eigenstates are no longer the same as the chirality eigenstates ψ_L and ψ_R .

By the way, the independent solutions to the free equations of motion for massless particles of any spin are the helicity eigenstates. For any spin, we will always find $\vec{S} \cdot \vec{p} \Psi_s = \pm s |\vec{p}| \Psi_s$, where $\vec{S} = \vec{J}$ are the rotation generators in the Lorentz group for spin s . For spin $\frac{1}{2}$, $\vec{S} = \frac{1}{2} \vec{\sigma}$. For spin 1, the rotation generators are given in Eqs. (10.14). For example, J_3 has eigenvalues ± 1 with eigenstates $(0, i, 1, 0)$ and $(0, -i, 1, 0)$. These are the states of circularly polarized light in the z direction, which are helicity eigenstates. In general, the polarizations of massless particles with spin > 0 can always be taken to be helicity eigenstates. This is true for spin $\frac{1}{2}$ and spin 1, as we have seen; it is also true for gravitons (spin 2). Rarita-Schwinger fields (spin $\frac{3}{2}$) and spins $s > 2$ (although, as we proved in Section 9.5.1, it is impossible to have interacting theories with massless fields of spin $s > 2$).

We have seen that the left- and right-handed chirality states ψ_L and ψ_R

- do not mix under Lorentz transformations -- they transform in separate irreducible representations.
- each have two components on which the $\vec{\sigma}$ -matrices act. These are the two spin states of the electron; both left- and right-handed spinors have two spin states.
- are eigenstates of helicity in the massless limit.

Spin is a vector quantity. We say spin up, or spin down, spin left, etc. It is the eigenvalue of $\vec{S} = \frac{\vec{\sigma}}{2}$ for a fermion. If there is no angular momentum, for example for a single particle, the spin and the rotation operators are identical $\vec{S} = \vec{J}$. We also talk about spin s as a scalar, which is the eigenvalue $s(s+1)$ of the operator $\vec{S}^2$. When we say spin $\frac{1}{2}$ we mean $s = \frac{1}{2}$.

Helicity refers to the projection of spin on the direction of motion. Helicity eigenstates satisfy

$\frac{\vec{S} \cdot \vec{p}}{|\vec{p}|} \Psi = \pm s \Psi$. Helicity eigenstates exist for any spin. The helicity eigenstates of the photon correspond to what we normally call circularly polarized light.

Chirality is a concept that only exists for spinors, or more precisely for (A, B) representations of the Lorentz group with $A \neq B$. You may remember the word chiral from chemistry: DNA is chiral, so is glucose and many organic molecules. These are not symmetric under reflection in a mirror. In field theory, a chiral theory is one that is not symmetric on interchange of the (A, B) representations with the (B, A) representations. Almost always, chirality means that a theory is not symmetric between left-handed

Weyl spinors ψ_L and right-handed spinors ψ_R . These chiral spinors can also be written as Dirac spinors that are eigenstates of γ_5 . By abuse of notation we also write ψ_L and ψ_R for Dirac spinors, with $\gamma_5\psi_L = -\psi_L$ and $\gamma_5\psi_R = \psi_R$. Whether a Weyl or Dirac spinor is meant by ψ_L and ψ_R will be clear from context. Chirality works for higher half-integer spins too. For example, a spin- $\frac{3}{2}$ field can be put in a Dirac spinor with a μ index, ψ_μ . Then $\gamma_5\psi_\mu = \pm\psi_\mu$ are the chirality eigenstates.

Whether spin, helicity or chirality is important depends on the physical question you are interested in. For free massless spinors, the spin eigenstates are also helicity eigenstates and chirality eigenstates. In other words, the Hamiltonian for the massless Dirac equation commutes with the operators for chirality, γ_5 , helicity, $\frac{\vec{S}\cdot\vec{p}}{E}$, and the spin operators, $\vec{S}$. The QED interaction $\bar{\psi}\gamma^\mu A_\mu\psi = \bar{\psi}_L\gamma^\mu A_\mu\psi_L + \bar{\psi}_R\gamma^\mu A_\mu\psi_R$ is non-chiral, that is, it preserves chirality. Helicity, on the other hand, is not necessarily preserved by QED: if a left-handed spinor has its direction reversed by an electric field, its helicity flips (Since the momentum of the spinor is flipped and the helicity is defined to be the projection of the spin onto the momentum, the helicity would also flip.). When particles are massless (or ultra-relativistic) they do not change direction so easily, but the helicity can flip due to an interaction.

In the massive case, it is also possible to take the non-relativistic limit. Then it is often better to talk about spin, the vector. Projecting on the direction of motion does not make so much sense when the particle is nearly at rest, or in a gas, say, when its direction of motion is constantly changing. The QED interactions do not preserve spin, however; only a strong magnetic field can flip an electron's spin. So, as long as magnetic fields are weak, spin is a good quantum number. That is why spin is used in quantum mechanics.

In QED, we hardly ever talk about chirality. The word is basically reserved for chiral theories, which are theories that are not symmetric under $L \leftrightarrow R$, such as the theory of the weak interactions. We talk very often about helicity. In the high-energy limit, helicity is often used interchangeably with chirality. As a slight abuse of terminology, we say ψ_L and ψ_R are helicity eigenstates. In the non-relativistic limit, we use helicity for photons and spin (the vector) for spinors. Helicity eigenstates for photons are circularly polarized light.

Solving the Dirac Equation

Now, we would like to solve the free Dirac equation. Since free spinors satisfy the Klein-Gordon equation, we can simply write down the solution:

$$\psi_s(x) = \int \frac{d^3p}{(2\pi)^3} u_s(p) e^{-ipx} \quad (8.1)$$

with $p_0 = \sqrt{\vec{p}^2 + m^2} > 0$ and $u_s(p)$ being the polarization vector for particles to be determined. These bases are dependent on momenta since there are no finite unitary representations of the Poincaré group. (Similarly to the spin-1 case.) Of course, we should also get solutions with negative energy. We now interpret these solutions as antiparticles, which has the same mass and opposite charge when coupling with the photon field. These solutions can be written as

$$\chi_s(x) = \int \frac{d^3p}{(2\pi)^3} v_s(p) e^{ipx} \quad (8.2)$$

with $p_0 = \sqrt{\vec{p}^2 + m^2} > 0$ and $v_s(p)$ being the polarization vector for antiparticles to be determined. These polarization vectors transform under the Poincaré group through the transformation of p^μ and the little group transformation that fixes p^μ . Thus, we only need to find explicit solutions at constant p^μ , as we did for spin-1 fields.

To begin, let us write the Dirac equation in the Weyl basis, which is

$$(i\gamma^\mu \partial_\mu - m)\psi = \begin{pmatrix} -m & i\sigma^\mu \partial_\mu \\ i\bar{\sigma}^\mu \partial_\mu & -m \end{pmatrix} \psi = 0. \quad (8.3)$$

Thus, the polarization vectors satisfy

$$\begin{pmatrix} -m & p \cdot \sigma \\ p \cdot \bar{\sigma} & -m \end{pmatrix} u_s(p) = \begin{pmatrix} -m & -p \cdot \sigma \\ -p \cdot \bar{\sigma} & -m \end{pmatrix} v_s(p) = 0. \quad (8.4)$$

In the rest frame, where $p^\mu = (m, 0, 0, 0)$, this becomes

$$\begin{pmatrix} -1 & 1 \\ 1 & -1 \end{pmatrix} u_s(p) = \begin{pmatrix} -1 & 1 \\ 1 & -1 \end{pmatrix} v_s(p) = 0, \quad (8.5)$$

which have the solutions

$$u_s = \begin{pmatrix} \xi_s \\ \xi_s \end{pmatrix}, \quad v_s = \begin{pmatrix} \eta_s \\ -\eta_s \end{pmatrix} \quad (8.6)$$

where ξ_s and η_s are constant two-component spinors. For example, we can choose ξ_s and η_s so that we get four linearly independent solutions:

$$u_\uparrow = \begin{pmatrix} 1 \\ 0 \\ 1 \\ 0 \end{pmatrix}, \quad u_\downarrow = \begin{pmatrix} 0 \\ 1 \\ 0 \\ 1 \end{pmatrix}, \quad v_\uparrow = \begin{pmatrix} -1 \\ 0 \\ 1 \\ 0 \end{pmatrix}, \quad v_\downarrow = \begin{pmatrix} 1 \\ 0 \\ -1 \\ 0 \end{pmatrix}, \quad (8.7)$$

which has four real degrees of freedom. The Dirac spinor is a complex four-component field with eight real degrees of freedom, and the equation of motion reduces it to four degrees of freedom.

More generally, consider the frame with momentum $p^\mu = (E, 0, 0, p_z)$, then

$$p \cdot \sigma = \begin{pmatrix} E - p_z & 0 \\ 0 & E + p_z \end{pmatrix}, \quad p \cdot \bar{\sigma} = \begin{pmatrix} E + p_z & 0 \\ 0 & E - p_z \end{pmatrix}. \quad (8.8)$$

One should notice that $\sqrt{E - p_z} \sqrt{E + p_z} = m$, and thus if we write $a = \sqrt{E - p_z}$ and $b = \sqrt{E + p_z}$ then (8.4) becomes

$$\begin{pmatrix} -ab & 0 & a^2 & 0 \\ 0 & -ab & 0 & b^2 \\ b^2 & 0 & -ab & 0 \\ 0 & a^2 & 0 & -ab \end{pmatrix} u_s(p) = \begin{pmatrix} -ab & 0 & -a^2 & 0 \\ 0 & -ab & 0 & -b^2 \\ -b^2 & 0 & -ab & 0 \\ 0 & -a^2 & 0 & -ab \end{pmatrix} v_s(p) = 0. \quad (8.9)$$

These have solutions

$$u_s = \begin{pmatrix} \begin{pmatrix} a & 0 \\ 0 & b \end{pmatrix} \xi_s \\ \begin{pmatrix} b & 0 \\ 0 & a \end{pmatrix} \xi_s \end{pmatrix}, \quad v_s = \begin{pmatrix} \begin{pmatrix} a & 0 \\ 0 & b \end{pmatrix} \eta_s \\ \begin{pmatrix} -b & 0 \\ 0 & -a \end{pmatrix} \eta_s \end{pmatrix}. \quad (8.10)$$

Using the simpler but slightly subtle notation (the square root of matrix can be defined by changing to the diagonal basis.)

$$\sqrt{p \cdot \sigma} = \begin{pmatrix} a & 0 \\ 0 & b \end{pmatrix}, \quad \sqrt{p \cdot \bar{\sigma}} = \begin{pmatrix} b & 0 \\ 0 & a \end{pmatrix}, \quad (8.11)$$

the above becomes

$$u_s = \begin{pmatrix} \sqrt{p \cdot \sigma} \xi_s \\ \sqrt{p \cdot \bar{\sigma}} \xi_s \end{pmatrix}, \quad v_s = \begin{pmatrix} \sqrt{p \cdot \sigma} \eta_s \\ -\sqrt{p \cdot \bar{\sigma}} \eta_s \end{pmatrix}. \quad (8.12)$$

In practice, we can always choose the momentum to point in the z -direction. Then, the four solutions are

$$u_p^1 = \begin{pmatrix} \sqrt{E - p_z} \\ 0 \\ \sqrt{E + p_z} \\ 0 \end{pmatrix}, \quad u_p^2 = \begin{pmatrix} 0 \\ \sqrt{E - p_z} \\ 0 \\ \sqrt{E + p_z} \end{pmatrix}, \quad v_p^1 = \begin{pmatrix} \sqrt{E - p_z} \\ 0 \\ -\sqrt{E + p_z} \\ 0 \end{pmatrix}, \quad v_p^2 = \begin{pmatrix} 0 \\ \sqrt{E - p_z} \\ 0 \\ -\sqrt{E + p_z} \end{pmatrix}. \quad (8.13)$$

In any frame of reference, u_s are always positive frequency electrons, and v_s are always negative frequency electrons, or equivalently, positive frequency positrons.

For massless spinors, $p_z = \pm E$ and the explicit solutions become 4-vectors with one non-zero component describing spinors with fixed helicity. The spinor solutions for massless electrons are sometimes called polarizations, and are useful for computing polarized electron scattering amplitudes.

For Weyl spinors, there are only four real degrees of freedom off-shell (not satisfying the equation of motion) and two real degrees of freedom on-shell (satisfying the equation of motion). Recalling that the Dirac equation splits up into separate equations for ψ_L and ψ_R , the Dirac spinors with zeros in the bottom two rows will be ψ_L and those with zeros in the top two rows will be ψ_R . Since ψ_L and ψ_R have two degrees of freedom each, these must be particle and antiparticle for the same helicity. The embedding of Weyl spinors into fields this way induces irreducible unitary representations of the Poincaré group for $m = 0$.

Normalisation and Outer Products

To find out what this implicit normalisation we have used in the above derivation is, consider the inner product

$$\begin{aligned}
\bar{u}_s(p)u_{s'}(p) &= \left(\frac{\sqrt{p \cdot \sigma} \xi_s}{\sqrt{p \cdot \bar{\sigma}} \xi_s} \right)^\dagger \gamma^0 \left(\frac{\sqrt{p \cdot \sigma} \xi_{s'}}{\sqrt{p \cdot \bar{\sigma}} \xi_{s'}} \right) \\
&= \left(\xi_s^\dagger \sqrt{p \cdot \bar{\sigma}} \quad \xi_s^\dagger \sqrt{p \cdot \sigma} \right) \begin{pmatrix} \sqrt{p \cdot \sigma} \xi_{s'} \\ \sqrt{p \cdot \bar{\sigma}} \xi_{s'} \end{pmatrix} \\
&= 2\xi_s^\dagger \sqrt{(p \cdot \bar{\sigma})(p \cdot \sigma)} \xi_{s'} \\
&= 2\xi_s^\dagger m \mathbb{1} \xi_{s'} \\
&= 2m\delta_{ss'}.
\end{aligned} \tag{8.14}$$

Similarly, we have

$$\bar{v}_s(p)v_{s'}(p) = -2m\delta_{ss'}. \tag{8.15}$$

Also, it is easy to show that

$$\bar{v}_s(p)u_{s'}(p) = \bar{u}_s(p)v_{s'}(p) = 0. \tag{8.16}$$

These are the conventional normalisations for the massive spinor inner product.

Of course, we can also compute

$$\begin{aligned}
u_s^\dagger(p)u_s(p) &= \left(\frac{\sqrt{p \cdot \sigma} \xi_s}{\sqrt{p \cdot \bar{\sigma}} \xi_s} \right)^\dagger \left(\frac{\sqrt{p \cdot \sigma} \xi_s}{\sqrt{p \cdot \bar{\sigma}} \xi_s} \right) \\
&= \left(\xi_s^\dagger \sqrt{p \cdot \sigma} \quad \xi_s^\dagger \sqrt{p \cdot \bar{\sigma}} \right) \begin{pmatrix} \sqrt{p \cdot \sigma} \xi_s \\ \sqrt{p \cdot \bar{\sigma}} \xi_s \end{pmatrix} \\
&= \xi_s^\dagger (p \cdot \sigma) \xi_s + \xi_s^\dagger (p \cdot \bar{\sigma}) \xi_s \\
&= 2\xi_s^\dagger E_p \mathbb{1} \xi_s \\
&= 2E_p \delta_{ss'},
\end{aligned} \tag{8.17}$$

and similarly

$$v_s^\dagger(p)v_s(p) = 2E_p \delta_{ss'}. \tag{8.18}$$

This is the conventional normalisation for massless Dirac spinors. Another useful relation is that

$$v_s^\dagger(p)u_s(\bar{p}) = u_s^\dagger(p)v_s(\bar{p}) = 0, \tag{8.19}$$

where $\bar{p}^\mu = (E_p, -\vec{p})$.

Now, let us compute the outer products. In quantum mechanics, we have $\sum |s\rangle \langle s|$. By analogy, we can define the outer product of spinor to be

$$\begin{aligned}
\sum_{s=1}^2 u_s(p) \bar{u}_s(p) &= \sum_{s=1}^2 \begin{pmatrix} \sqrt{p \cdot \sigma} \xi_s \\ \sqrt{p \cdot \bar{\sigma}} \xi_s \end{pmatrix} \begin{pmatrix} \sqrt{p \cdot \sigma} \xi_s \\ \sqrt{p \cdot \bar{\sigma}} \xi_s \end{pmatrix}^\dagger \begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix} \\
&= \sum_{s=1}^2 \begin{pmatrix} \sqrt{p \cdot \sigma} \xi_s \\ \sqrt{p \cdot \bar{\sigma}} \xi_s \end{pmatrix} (\xi_s^\dagger \sqrt{p \cdot \bar{\sigma}} \quad \xi_s^\dagger \sqrt{p \cdot \sigma}) \\
&= \sum_{s=1}^2 \begin{pmatrix} m & p \cdot \sigma \\ p \cdot \bar{\sigma} & m \end{pmatrix} \xi_s \xi_s^\dagger \\
&= \sum_{s=1}^2 (\not{p} + m) \xi_s \xi_s^\dagger.
\end{aligned} \tag{8.20}$$

Note that an arbitrary 2×2 matrix can be written as $\sum_{ss'} a_{ss'} \xi_s \xi_{s'}^\dagger$. Using our normalisation, we would find

$$\sum_{s=1}^2 \xi_s \xi_s^\dagger \sum_{ab=1}^2 a_{ab} \xi_a \xi_b^\dagger = \sum_{ab=1}^2 \sum_{s=1}^2 a_{ab} \xi_s \delta_{sa} \xi_b^\dagger = \sum_{ab=1}^2 a_{ab} \xi_a \xi_b^\dagger. \tag{8.21}$$

Thus, multiplying $\sum_s \xi_s \xi_s^\dagger$ on a matrix gives a factor of 1. Thus, we have

$$\sum_{s=1}^2 u_s(p) \bar{u}_s(p) = \not{p} + m, \tag{8.22}$$

and similarly

$$\sum_{s=1}^2 v_s(p) \bar{v}_s(p) = \not{p} - m. \tag{8.23}$$

Majorana Fermions

As discussed before, if the spinors are anticommuting Grassman numbers (which they indeed are), then we can write a Lagrangian with only one Weyl spinor with a mass term:

$$\mathcal{L} = i\psi_L^\dagger \sigma^\mu \partial_\mu \psi_L + i\frac{m}{2}(\psi_L^\dagger \sigma_2 \psi_L^* - \psi_L^T \sigma_2 \psi_L). \tag{9.1}$$

This Lagrangian describes the Majorana fermions, which transform under the same representation as the Weyl spinors. The difference comes in quantum theory, in which a Majorana fermion is its own antiparticle. Below, we will make this point clearer using the notion of charge conjugation.

Sometimes, it is easier to work with the Dirac algebra, so we can put the Majorana fermions into a Dirac four-component spinor:

$$\psi = \begin{pmatrix} \psi_L \\ i\sigma_2 \psi_L^* \end{pmatrix}. \tag{9.2}$$

Since $\sigma_2 \psi_L$ transforms just like ψ_R , the above transforms exactly like a Dirac spinor. Hence, we can write the Majorana mass as

$$\frac{m}{2} \bar{\psi} \psi = i \frac{m}{2} (\psi_L^\dagger \sigma_2 \psi_L^* - \psi_L^T \sigma_2 \psi_L). \quad (9.3)$$

Now, consider $-i\gamma^2 \psi^*$ under the Weyl basis:

$$-i\gamma^2 \psi^* = -i \begin{pmatrix} & \sigma_2 \\ -\sigma_2 & \end{pmatrix} \begin{pmatrix} \psi_L^* \\ -i\sigma_2^* \psi_L \end{pmatrix} = \begin{pmatrix} -\sigma_2^* \sigma \psi_L \\ i\sigma_2 \psi_L^* \end{pmatrix} = \begin{pmatrix} \psi_L \\ i\sigma_2 \psi_L^* \end{pmatrix} = \psi. \quad (9.4)$$

Therefore, the operator $-i\gamma^2$ takes the Majorana fermion to itself. What does this operator mean? Recall the Dirac equation

$$(i\not{\partial} - e\not{A} - m)\psi = 0. \quad (9.5)$$

Taking its complex conjugate, we find

$$(-i\gamma^{\mu*} \partial_\mu - e\gamma^{\mu*} A_\mu - m)\psi^* = 0. \quad (9.6)$$

Recall $(\gamma^2)^2 = -1$, the above implies

$$\gamma^2 (-i\gamma^{\mu*} \partial_\mu - e\gamma^{\mu*} A_\mu - m) \gamma^2 (-i\gamma^2 \psi^*) = 0, \quad (9.7)$$

where we have multiplied $-i\gamma^2$ on the right on both sides. Now, we can define a new representation for the γ -matrices $\hat{\gamma}^\mu = \gamma^2 \gamma^{\mu*} \gamma^2$, which also satisfies the Dirac algebra. (Trivial to check.) Thus, we find

$$\begin{aligned} & (-i\hat{\gamma}^\mu \partial_\mu - e\hat{\gamma}^\mu A_\mu + m)(-i\gamma^2 \psi^*) = 0 \\ \implies & (i\hat{\gamma}^\mu \partial_\mu + e\hat{\gamma}^\mu A_\mu - m)(-i\gamma^2 \psi^*) = 0, \end{aligned} \quad (9.8)$$

which means $-i\gamma^2 \psi^*$ satisfies the Dirac equation under a different representation of the γ matrix with the same mass and opposite charge. Therefore, the transformation takes the spinor to its charge conjugate. Thus, we define such a transformation as charge conjugation, C :

$$C : \quad \psi \rightarrow -i\gamma^2 \psi^* = \psi_c. \quad (9.9)$$

Thus, (9.4) says that the Majorana spinor is its own charge conjugate. Therefore, under a U(1) gauged or global symmetry theory, the Majorana spinor cannot be charged. Under such a symmetry $\psi \rightarrow e^{i\alpha} \psi$ and $\psi_c \rightarrow e^{-i\alpha} \psi_c$, so $\psi = \psi_c$ cannot hold. We can also see this through the mass term, which is not invariant under the U(1) transformation:

$$\psi_L^T \sigma_2 \psi_L \rightarrow \psi_L^T e^{i\alpha} \sigma_2 e^{i\alpha} \psi_L = e^{2i\alpha} \psi_L^T \sigma_2 \psi_L. \quad (9.10)$$

This is true for gauge charges, that is, those with a corresponding gauge boson, such as the photon, and also for global charges such as lepton number (which counts the number of electrons and neutrinos minus the number of positrons and antineutrinos), which have no associated gauge boson. If there are multiple Majorana fermions, they can transform together under a real representation of an internal non-Abelian symmetry group. For example, gluons in supersymmetry can be Majorana, transforming under the adjoint representation of SU(3).

There are particles in nature called neutrinos, which apparently carry no charges. Thus, they may be Majorana or Dirac fermions. In fact, a number of experiments are trying hard to find out if neutrinos are Majorana. Weyl spinors do exist in nature, in an obvious way, since Dirac spinors are just two Weyl spinors put together. But Weyl spinors are also integral to the theory of weak interactions, which is chiral.

Charge Conjugation

To bring our understanding of the notion of charge conjugation a bit further, let us consider what would happen if we act the charge conjugation operation on the basis of the free Dirac spinors. This gives

$$(u_{\uparrow})^c = -i\gamma^2 u_{\uparrow}^* = -i \begin{pmatrix} 0 & 0 & 0 & -i \\ 0 & 0 & i & 0 \\ 0 & i & 0 & 0 \\ -i & 0 & 0 & 0 \end{pmatrix} \begin{pmatrix} 1 \\ 0 \\ 1 \\ 0 \end{pmatrix} = \begin{pmatrix} 0 \\ 1 \\ 0 \\ -1 \end{pmatrix} = v_{\downarrow}. \quad (10.1)$$

Similarly, we have

$$(u_{\downarrow})^c = v_{\uparrow}, \quad (u_{\uparrow})^c = v_{\downarrow}, \quad (v_{\uparrow})^c = u_{\downarrow}, \quad (v_{\downarrow})^c = u_{\uparrow}. \quad (10.2)$$

Therefore, charge conjugation takes particles to antiparticles and flips their spins.

Charge conjugation may or may not be a symmetry of a particular Lagrangian. The operation of charge conjugation acts on spinors and their conjugates by

$$C : \psi \rightarrow -i\gamma_2 \psi^*. \quad (10.3)$$

In the Weyl basis, $\gamma_2^* = -\gamma_2$ and $\gamma_2^T = \gamma_2$, so

$$C : \psi^* \rightarrow -i\gamma_2 \psi, \quad (10.4)$$

and in particular $C^2 = 1$, which is why C is called a *conjugation* operator. Then

$$C : \bar{\psi}\psi \rightarrow (-i\gamma_2 \psi)^T \gamma_0 (-i\gamma_2 \psi^*) = -\psi^T \gamma_2^T \gamma_0 \gamma_2 \psi^* = -\psi^T \gamma_0 \psi^*. \quad (10.5)$$

The transpose on a spinor is not really necessary. This last expression just means

$$-\psi^T \gamma_0 \psi^* = -(\gamma_0)_{\alpha\beta} \psi_{\alpha} \psi_{\beta}^*. \quad (10.6)$$

Now, anticommuting the spinors, relabeling $\alpha \leftrightarrow \beta$ and combining shows that

$$-(\gamma_0)_{\alpha\beta} \psi_{\alpha} \psi_{\beta}^* = (\gamma_0)_{\alpha\beta} \psi_{\beta}^* \psi_{\alpha} = (\gamma_0)_{\beta\alpha} \psi_{\alpha}^* \psi_{\beta} = \psi^{\dagger} \gamma_0^T \psi = \bar{\psi}\psi. \quad (10.7)$$

Thus,

$$C : \bar{\psi}\psi \rightarrow \bar{\psi}\psi. \quad (10.8)$$

Similarly,

$$C : \bar{\psi}\gamma^\mu\partial_\mu\psi \rightarrow \bar{\psi}\gamma^\mu\partial_\mu\psi. \quad (10.9)$$

So the free Dirac Lagrangian is C invariant.

We can also check that

$$C : \bar{\psi}\gamma^\mu\psi \rightarrow -\bar{\psi}\gamma^\mu\psi. \quad (10.10)$$

This implies that the interaction $eA_\mu\bar{\psi}\gamma^\mu\psi$ will only be C invariant if

$$C : A_\mu \rightarrow -A_\mu. \quad (10.11)$$

Since the kinetic term $F_{\mu\nu}^2$ is invariant under $A_\mu \rightarrow \pm A_\mu$, the whole QED Lagrangian is therefore C invariant.

The transformation $A_\mu \rightarrow -A_\mu$ under C may seem strange, since a vector field is real, so it should not transform under an operation that switches particles with antiparticles. Since particles and antiparticles have opposite charge and A_μ couples proportionally to charge, this transformation is needed to compensate for the transformation of the charged fields.

There is an important lesson here: you could take $C : A_\mu \rightarrow A_\mu$, but then the Lagrangian would not be invariant. Thus, rather than trying to figure out how C acts, the right question is: How can we enlarge the action of the transformation C , which we know for Dirac spinors, to a full interacting theory so that the symmetry is preserved? Whether we interpret C with the words “takes particles to antiparticles”, has no physical implications. In contrast, a symmetry of a theory does have physical implications: preservation of the symmetry gives a superselection rule - certain transitions cannot happen. An important example is that C invariance forces matrix elements involving an odd number of photons to vanish, a result known as Furry's theorem. Thus, cataloging the symmetries of a theory is important, whether or not we have interesting names or simple physical interpretations of those symmetries.

Let us prove some useful transformations for future reference:

$$\begin{aligned} C : \quad i\bar{\psi}\gamma^5\psi &\rightarrow i(-i\gamma^2\psi)^T\gamma^0\gamma^5(-i\gamma^2\psi^*) \\ &= -i\psi(\gamma^2)^T\gamma^0\gamma^5\gamma^2\psi^* \\ &= i\psi^T\gamma^0\gamma^5\gamma^2\gamma^2\psi^* \\ &= -i\psi^T\gamma^0\gamma^5\psi^* \\ &= -i\psi_\alpha\gamma_{\alpha\beta}^0\gamma_{\beta\gamma}^5\psi_\gamma^* \\ &= i\psi_\gamma^*\gamma_{\gamma\beta}^0\gamma_{\beta\alpha}^5\psi_\alpha \\ &= i\bar{\psi}\gamma^5\psi. \end{aligned} \quad (10.12)$$

$$\begin{aligned} C : \quad i\bar{\psi}\gamma^5\gamma^\mu\psi &\rightarrow i(-i\gamma^2\psi)^T\gamma^0\gamma^5\gamma^\mu(-i\gamma^2\psi^*) \\ &= -i\psi(\gamma^2)^T\gamma^0\gamma^5\gamma^\mu\gamma^2\psi^* \\ &= i\psi^T\gamma^0\gamma^5(2g^{\mu 2} - \gamma^\mu\gamma^2)\gamma^2\psi^*. \end{aligned} \quad (10.13)$$

Note, when $\mu \neq 2$, we find

$$i\psi^T\gamma^0\gamma^5(2g^{\mu 2} - \gamma^\mu\gamma^2)\gamma^2\psi^* = -i\psi^T\gamma^0\gamma^5\gamma^\mu\psi^* = i\psi^\dagger\gamma^0\gamma^5\gamma^\mu\psi. \quad (10.14)$$

Then, when $\mu = 2$, we have

$$i\psi^T \gamma^0 \gamma^5 (2 \cancel{\gamma^2 \gamma^2}^{-1} - \cancel{\gamma^2 \gamma^2}^{-1} \gamma^2 \psi^* = -i\psi^T \gamma^0 \gamma^5 \gamma^2 \psi^* = i\psi^\dagger \gamma^0 \gamma^5 \gamma^2 \psi. \quad (10.15)$$

Thus, we can conclude

$$C : \quad i\bar{\psi} \gamma^5 \gamma^\mu \psi \rightarrow i\bar{\psi} \gamma^5 \gamma^\mu \psi. \quad (10.16)$$

In the above derivation, we can see that since γ^5 commutes with all the rest of γ -matrices, it gives a factor of 1 while taking charge conjugation. Now, if we put in another γ^ν , we would need to swap γ^2 with γ^μ twice, which results in an extra factor of -1 in contrast to (10.16). Thus,

$$C : \quad i\bar{\psi} \gamma^\mu \gamma^\nu \psi \rightarrow -i\bar{\psi} \gamma^\mu \gamma^\nu \psi, \quad (10.17)$$

and thus

$$C : \quad i\bar{\psi} \sigma_{\mu\nu} \psi \rightarrow -i\bar{\psi} \sigma_{\mu\nu} \psi \quad (10.18)$$

Parity

Recall that the full Lorentz group, $O(1, 3)$, is the group of 4×4 matrices Λ with $\Lambda^T g \Lambda = g$. In addition to the transformations smoothly connected to $\mathbb{I}$, this group also contains the transformations of parity and time reversal:

$$P : (t, \vec{x}) \rightarrow (t, -\vec{x}), \quad (11.1)$$

$$T : (t, \vec{x}) \rightarrow (-t, \vec{x}). \quad (11.2)$$

Just as with charge conjugation, we would like to know how to define these transformations acting on spinors and other fields, so that they are symmetries of QED or whatever theory we are studying.

You might expect that the action of P and T should be determined from representation theory.

However, recall that technically spinors do not actually transform under the Lorentz group, $O(1, 3)$, only its universal cover, $SL(2, \mathbb{C})$, so we are not guaranteed that T and P will act in any nice way on irreducible spinor representations. In fact, they do not. Although we can define an action of T and P on spinors (and other fields), these definitions are only useful to the extent that they are symmetries of the theory we are interested in. For example, we will define P so that it is a good symmetry of QED, but there is no way to define it so that it is preserved under the weak interactions. In any representation, we should have $P^2 = T^2 = 1$.

Scalars and Vectors

For real scalar fields, parity should be symmetry for the kinetic term $\mathcal{L} = -\frac{1}{2} \phi \square \phi - \frac{1}{2} m^2 \phi^2$. Thus, with $P^2 = 1$, we are only left with two choices:

$$P : \quad \phi(t, \vec{x}) \rightarrow \pm \phi(t, -\vec{x}). \quad (11.3)$$

The sign is known as the intrinsic parity. In nature, there are particles with even parity (scalar, such as the Higgs bosons) and particles with odd parity (pseudoscalars, such as the π^0). Since the action integrates over all $\vec{x}$, the action is invariant under $\vec{x} \rightarrow -\vec{x}$.

For complex scalar fields, the Lagrangian is $\mathcal{L} = -\phi^* \square \phi - m^2 \phi^* \phi$. Thus, the most general parity transformation would be

$$P : \quad \phi(t, \vec{x}) \rightarrow \eta \phi(t, -\vec{x}), \quad (11.4)$$

where η is a pure phase. Recall, any charged scalar must have a global symmetry under the phase transformation $\phi \rightarrow e^{i\alpha} \phi$, which is why they can couple to the photon. Therefore, η can be combined with a phase shift, which means it is not well defined. Thus, we can choose a phase convention for one charged particle, and all the other phase will become physical. Hence, we redefine P so that all particles have parity phases ± 1 . In specific, we choose proton, neutron and electron to have parity $+1$, and all the rest to have parity -1 .

For the free theory of vector fields to be invariant under parity, we require

$$P : \quad V_0(t, \vec{x}) \rightarrow \pm V_0(t, -\vec{x}), \quad V_i(t, \vec{x}) \rightarrow \mp V_i(t, -\vec{x}). \quad (11.5)$$

If the fields transform like $\vec{x}$, $V_i \rightarrow -V_i$, we call it a vector with parity -1 . If the fields transform like a scalar, $V_i \rightarrow V_i$, we call it a pseudovector with parity $+1$.

Massless vectors such as the photon have to have parity -1 . To see this, just look at the coupling to a charged scalar. Under parity we would like

$$P : \quad A_\mu(\phi^* \partial_\mu \phi - \phi \partial_\mu \phi^*) \rightarrow A_\mu(\phi^* \partial_\mu \phi - \phi \partial_\mu \phi^*), \quad (11.6)$$

which is only possible if A_μ transforms like ∂_μ . That is, A_μ is a vector:

$$P : \quad A_0(t, \vec{x}) \rightarrow A_0(t, -\vec{x}), \quad A_i(t, \vec{x}) \rightarrow -A_i(t, -\vec{x}). \quad (11.7)$$

Spinors

Now, let us consider the spinors. In the Lorentz group, parity commutes with rotation, which means it should not change the spin of a state embedded in a vector field. For massless spinors, recall that the left- and right-handed spinors are eigenstates of the helicity operator, which projects spin onto the momentum axis:

$$\frac{\vec{\sigma} \cdot \vec{p}}{|\vec{p}|} \psi_R = \psi_R, \quad \frac{\vec{\sigma} \cdot \vec{p}}{|\vec{p}|} \psi_L = -\psi_L. \quad (11.8)$$

Since parity commutes with spin and energy, but flips momentum, it will take a left-handed spinor to a right-handed one and vice versa. In other words, it maps $(\frac{1}{2}, 0)$ representations to $(0, \frac{1}{2})$. Therefore, parity cannot be appended to any of the irreducible spin- $\frac{1}{2}$ representation alone.

For the Dirac spinor, under the Weyl basis, we can simply write

$$P : \quad \psi \rightarrow \gamma^0 \psi. \quad (11.9)$$

Similarly to the complex scalar case, we have a phase ambiguity here. However, we have chosen parity $+1$ as a convention, using global gauge invariance just as before. Note, a chiral theory (no symmetry under $L \leftrightarrow R$) is not invariant under parity, such as the theory of weak interactions

Now, note that

$$P : \quad \bar{\psi}\psi(t, \vec{x}) \rightarrow \psi^\dagger \gamma^0 \gamma^0 \gamma^0 \psi(t, -\vec{x}) = \bar{\psi}\psi(t, -\vec{x}), \quad (11.10)$$

$$P : \quad \bar{\psi}\gamma^\mu \psi(t, \vec{x}) \rightarrow \psi^\dagger \gamma^0 \gamma^0 \gamma^\mu \gamma^0 \psi(t, -\vec{x}) = \bar{\psi}\gamma^{\mu\dagger} \psi(t, -\vec{x}). \quad (11.11)$$

Recall, $\gamma_0^\dagger = \gamma_0$ and $\gamma_i^\dagger = -\gamma_i$, and thus

$$P : \quad \bar{\psi}\gamma^0 \psi(t, \vec{x}) \rightarrow \bar{\psi}\gamma^0 \psi(t, -\vec{x}), \quad \bar{\psi}\gamma^i \psi(t, \vec{x}) \rightarrow -\bar{\psi}\gamma^i \psi(t, -\vec{x}), \quad (11.12)$$

which transforms just as a four-vector. Therefore, $\bar{\psi}\not{\partial}\psi$, $\bar{\psi}\not{A}\psi$ and $\bar{\psi}\psi$ is invariant under parity, and so is the Dirac Lagrangian. Things get a bit different if we have an extra γ^5 :

$$P : \quad \bar{\psi}\gamma^0 \gamma^5 \psi(t, \vec{x}) \rightarrow -\bar{\psi}\gamma^0 \gamma^5 \psi(t, -\vec{x}), \quad \bar{\psi}\gamma^i \gamma^5 \psi(t, \vec{x}) \rightarrow \bar{\psi}\gamma^i \gamma^5 \psi(t, -\vec{x}). \quad (11.13)$$

This leads to

$$P : \quad \bar{\psi}\not{A}\psi(t, \vec{x}) \rightarrow \bar{\psi}\not{A}\psi(t, -\vec{x}), \quad (11.14)$$

$$P : \quad \bar{\psi}\not{A}\gamma^5 \psi(t, \vec{x}) \rightarrow -\bar{\psi}\not{A}\gamma^5 \psi(t, -\vec{x}). \quad (11.15)$$

The currents contracted with A^μ in these terms are known as the vector current, $J^\mu = \bar{\psi}\gamma^\mu \psi$, and the axial vector current, $J_5^\mu = \bar{\psi}\gamma^\mu \gamma^5 \psi$. These currents play a crucial role in the theory of weak interactions, which involves $J^\mu - J_5^\mu$, or the V--A current.

Time Reversal

Finally, let us consider time reversal. As a Lorentz transformation,

$$T : \quad (t, \vec{x}) \rightarrow (-t, \vec{x}). \quad (12.1)$$

We require the kinetic term of the spinor fields to be invariant under T , which means we need $\bar{\psi}\gamma^\mu \psi$ to transform as a four-vector. Thus, we need

$$T : \quad \bar{\psi}\gamma^0 \psi(t, \vec{x}) \rightarrow -\bar{\psi}\gamma^0 \psi(-t, \vec{x}), \quad (12.2)$$

which is really strange. It says we need to turn a positive definite quantity ($\psi^\dagger \psi$) to a negative definite quantity ($-\psi^\dagger \psi$). This is impossible for any linear transformation $\psi \rightarrow \Gamma\psi$.

The Simple $\hat{T}$

We need to think harder. One way out would be to try $\psi \rightarrow \Gamma\psi^*$, so that

$$\begin{aligned}
\hat{T}: \quad \psi^\dagger \psi(t, \vec{x}) &\rightarrow (\Gamma \psi^*)^\dagger \Gamma \psi^*(-t, \vec{x}) \\
&= \psi^T \Gamma^\dagger \Gamma \psi^*(-t, \vec{x}) \\
&= \psi_\alpha \Gamma_{\alpha\beta}^\dagger \Gamma_{\beta\gamma} \psi_\gamma^*(-t, \vec{x}) \\
&= -\psi_\gamma^* \Gamma_{\alpha\beta}^\dagger \Gamma_{\beta\gamma} \psi_\alpha(-t, \vec{x}) \\
&= -\psi^\dagger (\Gamma^\dagger \Gamma)^T \psi(-t, \vec{x}).
\end{aligned} \tag{12.3}$$

Thus, we require $\Gamma^\dagger \Gamma = \mathbb{1}$, which means Γ needs to be unitary. However, we also need the transformation to preserve $\bar{\psi} \gamma^i \psi$ and the mass term $\bar{\psi} \psi$. For the mass term,

$$\hat{T}: \quad \bar{\psi} \psi \rightarrow \psi^T \Gamma^\dagger \gamma^0 \Gamma \psi^* = -\psi^\dagger (\Gamma^\dagger \gamma^0 \Gamma)^T \psi = -\bar{\psi} (\Gamma^\dagger \gamma^0 \Gamma \gamma^0)^T \psi, \tag{12.4}$$

which is equal to $\bar{\psi} \psi$ only when $\{\Gamma, \gamma^0\} = 0$. Then,

$$\hat{T}: \quad \bar{\psi} \gamma^i \psi \rightarrow \psi^T \Gamma^\dagger \gamma^0 \gamma^i \Gamma \psi^* = -\psi^\dagger (\Gamma^\dagger \gamma^0 \gamma^i \Gamma)^T \psi = -\bar{\psi} (\Gamma^\dagger \gamma^0 \gamma^i \Gamma \gamma^0)^T \psi = -\bar{\psi} (\Gamma^\dagger \gamma^i \Gamma)^T \psi, \tag{12.5}$$

which means we need

$$-(\Gamma^\dagger \gamma^i \Gamma)^T = \gamma^i \implies \gamma^i \Gamma + \Gamma \gamma^{i*} = 0. \tag{12.6}$$

This implies $[\Gamma, \gamma^1] = 0$, $\{\Gamma, \gamma^2\} = 0$ and $[\Gamma, \gamma^3] = 0$, which has the unique solution (up to a constant)

$$\Gamma = \gamma^0 \gamma^2, \tag{12.7}$$

and thus

$$\hat{T}: \quad \psi(t, \vec{x}) \rightarrow \gamma^0 \gamma^2 \psi^*(t, -\vec{x}), \quad \psi^\dagger(t, \vec{x}) \rightarrow -\psi^T \gamma^0 \gamma^2(-t, \vec{x}). \tag{12.8}$$

This is very similar to PC , where we had

$$PC: \quad A_0(t, \vec{x}) \rightarrow -A_0(-t, \vec{x}), \quad A_i(t, \vec{x}) \rightarrow A_i(t, -\vec{x}), \tag{12.9}$$

so that the interaction $\bar{\psi} \not{A} \psi$ is conserved.

Now consider the CPT transformation:

$$CPT: \quad \psi(t, \vec{x}) \rightarrow -i\gamma^2 (\gamma^0 [\gamma^0 \gamma^2 \psi^*])^*(-t, -\vec{x}) = -i\psi(-t, -\vec{x}), \tag{12.10}$$

and so

$$CPT: \quad \bar{\psi}(t, \vec{x}) \gamma^\mu \psi(t, \vec{x}) \rightarrow \bar{\psi}(-t, -\vec{x}) \gamma^\mu \psi(-t, -\vec{x}), \tag{12.11}$$

$$CPT: \quad A_\mu(t, \vec{x}) \rightarrow A_\mu(-t, -\vec{x}). \tag{12.12}$$

CPT also sends $\partial_\mu \rightarrow -\partial_\mu$. Thus, $\bar{\psi} \psi$, $i\bar{\psi} \gamma^\mu \partial_\mu \psi$ and $\bar{\psi} A_\mu \gamma^\mu \psi$ are all invariant in the action.

This time reversal is essentially defined to be $\hat{T} \sim (CP)^{-1}$, which makes CPT invariance trivial. The actual CPT theorem concerns a different T symmetry, which we will now discuss.

The Wigner's T

What we normally call time reversal is a symmetry T that was described by Wigner, which explains the Kramer's degeneracy. Let us first recap what is Kramer's degeneracy. Consider the Schrödinger equation

$$i\partial_t\psi(t, \vec{x}) = H\psi(t, \vec{x}), \quad (12.13)$$

where we will assume, for simplicity, that H is real and independent of time. If we take the complex conjugate of the above and $t \rightarrow -t$, we find

$$i\partial_t\psi^*(-t, \vec{x}) = H\psi^*(-t, \vec{x}). \quad (12.14)$$

Thus, $\psi^*(-t, \vec{x})$ is another solution to the Schrödinger equation. Therefore, if $\psi \neq \xi\psi^*$ for any complex number, then ψ^* will be another state with the same energy. The doubling state at each energy is known as Kramer's degeneracy. This theorem also holds for more complex systems.

The trick to Wigner's T is that we need to take the complex conjugate and then take $\psi \rightarrow \psi^*$. This means under a T transformation

$$T : \quad i \rightarrow -i, \quad \psi(t, \vec{x}) \rightarrow \tilde{\Gamma}\psi(-t, \vec{x}), \quad (12.15)$$

which says that T is an anti-linear operator. ($T(\psi_1 + i\psi_2) = T(\psi_1) - iT(\psi_2)$.)

Note, since T changes the sign of all i in the Lagrangian, it also effects the γ -matrices. In the Weyl basis, we find

$$T : \quad \gamma^2 \rightarrow -\gamma^2. \quad (12.16)$$

To preserve the kinetic term of the Lagrangian, $i\bar{\psi}\gamma^\mu\partial_\mu\psi$, we need $\bar{\psi}\gamma^0\psi \rightarrow \bar{\psi}\gamma^0\psi$ and $\bar{\psi}\gamma^i\psi \rightarrow -\bar{\psi}\gamma^i\psi$.

$$T : \quad \bar{\psi}\gamma^0\psi \rightarrow \psi^\dagger\tilde{\Gamma}^\dagger\gamma^0\tilde{\Gamma}\psi, \quad (12.17)$$

which implies

$$[\tilde{\Gamma}, \gamma^0] = 0. \quad (12.18)$$

Similarly, we find

$$\{\tilde{\Gamma}, \gamma^1\} = [\tilde{\Gamma}, \gamma^2] = \{\tilde{\Gamma}, \gamma^3\} = 0. \quad (12.19)$$

The only element of the Dirac algebra that satisfies these conditions $\tilde{\Gamma} = \gamma^1\gamma^3$, up to a constant. Thus, we take

$$T : \psi(t, \vec{x}) \rightarrow \gamma_1\gamma_3\psi(-t, \vec{x}) = \begin{pmatrix} 0 & 1 & 0 & 0 \\ -1 & 0 & 0 & 0 \\ 0 & 0 & 0 & 1 \\ 0 & 0 & -1 & 0 \end{pmatrix} \psi(-t, \vec{x}). \quad (12.20)$$

Thus, T flips the spins of particles, but does not turn particles into antiparticles, as expected. T does not have a well-defined action on Weyl spinors, which have one spin state. T also reverses the momenta, $\vec{p} = i\vec{\nabla}$, because of the i . Thus, T makes it look like things are going forwards in time, but with their momenta and spins flipped (just like playing a movie backward).

Similarly, for $\bar{\psi}\not{A}\psi$ to be invariant, we need A_μ to transform as $i\partial_\mu$, which is

$$T : \quad A_0(t, \vec{x}) \rightarrow A_0(-t, \vec{x}), \quad A_i(t, \vec{x}) \rightarrow -A_i(-t, \vec{x}). \quad (12.21)$$

It is straightforward to check now that the Dirac Lagrangian is invariant under T .

Next, consider the combined operation of CPT . This sends particles into antiparticles moving as if you watched them in reverse in a mirror. On Dirac spinors, it acts as

$$C \cdot P \cdot T : \quad \psi(x) \rightarrow -i\gamma_2\gamma_0\gamma_1\gamma_3\psi^*(-x) = -\gamma_5\psi^*(-x). \quad (12.22)$$

It also sends $A_\mu(x) \rightarrow A_\mu(-x)$, $\phi \rightarrow \phi^*(-x)$ and of course $i \rightarrow -i$.

Any terms you could possibly write down, for example,

$$\mathcal{L} = \bar{\psi}\psi, \quad i\bar{\psi}\phi\psi, \quad i\bar{\psi}\not{A}\psi, \quad \bar{\psi}\gamma^\mu\gamma^5\psi W_\mu, \quad i\bar{\psi}\sigma^{\mu\nu}\psi F_{\mu\nu} \quad (12.23)$$

and so on, are all CPT invariant. The CPT theorem says that this is a consequence of Lorentz invariance and unitarity.